CAREDOSE: SMART MEDICINE REMINDER & HEALTH COMPANION

A MINI PROJECT REPORT

Submitted by

GAYATHRI KRISHNAN

MLM25MCA-2030

To

the APJ Abdul Kalam Technological University in
Partial fulfilment of the requirements for the award
of the degree

of

MASTER OF COMPUTER APPLICATIONS

Under the Guidance of

Ms. ANJALY SAJEEV

(Assistant Professor, Department of Computer Applications)

DEPARTMENT OF COMPUTER APPLICATIONS

MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR

(Affiliated to APJ Abdul Kalam Technological University)

OCTOBER 2026

MANGALAM COLLEGE OF ENGINEERING

Accredited by NAAC& ISO 9001:2000 Certified Institution

DEPARTMENT OF COMPUTER APPLICATIONS

VISION

To become a centre of excellence in computer applications,competent in the global ecosystem with technical knowledge,innovation with a sense of social commitment.

MISSION

- To serve with state of the art education,foster advanced research and cultivate innovation in the field of computer applications.
- To prepare learners with knowledge skills and critical thinking to excel in the technological landscape and contribute positively to society.

Program Educational Objectives

- PEO I :Graduates will possess a solid foundation and in-depth understanding of computer applications and will be equipped to analyze real-world problems, design and create innovative solutions, and effectively manage and maintain these solutions in their professional careers.
- PEO II: Graduates will acquire technological advancements through continued education, lifelong learning and research, thereby making meaningful contributions to the field of computing.
- PEO III: Graduates will cultivate team spirit, leadership, communication skills, ethics, and social values, enabling them to apply their understanding of the societal impacts of computer applications effectively.

Program Specific Outcomes

- **PSO I**: Apply advanced technologies through innovations to enhance the efficiency of design and development.
- **PSO II**: Apply the principles of computing to analyze, design and implement sustainable solutions for real world challenges.

MAPPING OF PO-PSO-SDG

1. MAPPING WITH PROGRAM OUTCOMES (POs):

SL.NO	POs ADDRESSED	RELEVANCE TO PROJECT
1	PO1	Applies programming, database, and web development concepts to build CareDose.
2	PO2	Analyzes medication management, health tracking, and caretaker support challenges.
3	PO3	CareDose develops an integrated solution for medicine scheduling, tracking, and health records.
4	PO4	Uses HTML, CSS, JavaScript, Python, Django REST Framework, and SQLite.
5	PO5	Demonstrates planning, development, testing, debugging, and documentation skills.
6	PO6	The system follows a modular development approach covering requirements, design, implementation, testing, database management and documentation.

LIST OF PROGRAM OUTCOMES (POs):

PO1 (**Foundation Knowledge):** Apply knowledge of programming logic, database concepts and coding fundamentals for solution architecture and problem solving.

PO2 (Problem Analysis): Identify, review, formulate and analyze problems in medication management with focus on user requirements.

PO3 (Development of Solutions): Design and develop an integrated medication-management solution for efficient medicine scheduling, tracking and health-information organization.

PO4 (Modern Tool Usage): Select and apply modern computational tools and technologies such as HTML, CSS, JavaScript, Python, Django REST Framework, OCR and SQLite for system development.

PO5 (**Individual and Teamwork**)**:** Function and communicate effectively as an individual or team member during the planning, development and testing of the project.

PO6 (**Project Management and Finance):** Apply project management principles such as planning, scheduling, resource management and work breakdown structure.

2. MAPPING WITH PROGRAM SPECIFIC OUTCOMES (PSOs):

LIST OF PROGRAM SPECIFIC OUTCOMES (PSOs):

PSO 1: Apply advanced technologies through innovations to enhance the efficiency of design development.

PSO 2: Apply the principles of computing to analyze, design and implement sustainable solutions for real world challenges.

SL.NO	PSOs ADDRESSED	RELEVANCE TO PROJECT
1	PSO1 – Apply advanced technologies through innovations to enhance the efficiency of design development	HTML, CSS, JavaScript, Python, Django REST Framework, OCR, QR-based information access and visualization are combined to develop an organized digital health-support platform.
2	PSO2 – Apply the principles of computing to analyze, design and implement sustainable solutions for real-world challenges	Computing principles of database design, authentication, access control, REST APIs and modular software engineering are applied to address practical medication-management needs.

3. MAPPING WITH SUSTAINABLE DEVELOPMENT GOALS (SDGs):

SDG NO	SDGs ADDRESSED	RELEVANCE TO PROJECT
SDG 3	Good Health and Well-Being	CareDose supports organized medication schedules, medication records, health-record maintenance and access to selected essential health information.
SDG 9	Industry, Innovation and Infrastructure	The project applies web technologies, REST APIs, OCR, QR-based access and data visualization to create a digital healthcare-support platform.
SDG 10	Reduced Inequalities	A simple centralized interface and caretaker support can make medication-management assistance more accessible to users who need additional support.

SUSTAINABLE DEVELOPMENT GOALS (SDGs):

SDG 3 - Good Health and Well-Being: Ensure healthy lives and promote well-being for all at all ages.

SDG 9 - Industry, Innovation and Infrastructure: Build resilient infrastructure, promote inclusive and sustainable industrialisation and foster innovation.

SDG 10 - Reduced Inequalities: Reduce inequality within and among countries.

MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR

DEPARTMENT OF COMPUTER APPLICATIONS

OCTOBER 2026



DECLARATION

I hereby certify that the work presented in the project entitled "CAREDOSE: SMART MEDICINE REMINDER & HEALTH COMPANION" submitted to the DEPARTMENT OF COMPUTER APPLICATIONS is an authentic record of my own work carried out under the supervision of MS. ANJALY SAJEEV, Assistant Professor. This work has not been submitted to any other institution or university for the award of any other degree. The report has been checked for plagiarism by the college and the similarity index is within the permissible limit set by the college.

Date:

Place:

Name & Signature of Student

GAYATHRI KRISHNAN

MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR
DEPARTMENT OF COMPUTER APPLICATIONS
OCTOBER 2026



CERTIFICATE

Certified that the report entitled "**CAREDOSE: SMART MEDICINE REMINDER & HEALTH COMPANION**" submitted by **GAYATHRI KRISHNAN MLM25MCA-2030** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the **DEGREE OF MASTER OF COMPUTER APPLICATIONS** by APJ **ABDUL KALAM TECHNOLOGICAL UNIVERSITY** Is a bonafide record of the **MINI PROJECT** work carried out by her under our guidance and supervision during the academic year 2025-27.

PROJECT GUIDE
Ms. Anjaly Sajeev
Assistant Professor
Department of Computer Applications

PROJECT COORDINATOR
Ms. Athira R Kurup
Assistant Professor
Department of Computer Application

HEAD OF THE DEPARTMENT
Ms. Divya S B
HOD & Assistant Professor
Department of Computer Applications

INTERNAL EXAMINER

ACKNOWLEDGEMENT

I am greatly indebted to the authorities of Mangalam College of Engineering for providing the necessary facilities to successfully complete my Project on the topic "CAREDOSE: Smart Medicine Reminder & Health Companion".

I express my sincere thanks to **Dr. Vinodh P Vijayan**, Principal, Mangalam College of Engineering, for providing the facilities and support to complete my Project successfully.

I thank and express my sincere gratitude to **Ms. Divya S B**, HOD & Assistant Professor, Department of Computer Applications, Mangalam College of Engineering, for her invaluable help and support, which helped me a lot in successfully completing this Project work.

I express my gratitude to my Internal Guide, **Ms. Anjaly Sajeev**, Assistant Professor, Department of Computer Applications, for her valuable suggestions, guidance, and encouragement, which helped in the successful completion of my Project.

Furthermore, I would like to acknowledge with much appreciation the crucial role of the faculties, especially Project Coordinator, **Ms. Athira R Kurup**, Department of Computer Applications, Mangalam College of Engineering, who gave me the necessary guidance, permission, and resources required to complete the project presentation and report.

Finally, I would like to express my heartfelt thanks to my parents for their continuous support, encouragement, and motivation throughout the completion of this Project.

GAYATHRI KRISHNAN

MLM25MCA-2030

ABSTRACT

CareDose: Smart Medicine Reminder & Health Companion is a web-based healthcare-support system designed to help users organize medicines, track medication intake, maintain personal health information, and share selected information with a trusted caretaker. The project addresses the practical difficulty of remembering medicines, maintaining medication records, and keeping important health details available when required.

The system provides user registration and authentication, personal profile management, medicine and prescription management, medication reminders and tracking, medication history, health-record management, caretaker access, an emergency health QR facility, and an administrative interface. Medicine records can include medicine name, dosage, timing, food timing, frequency, duration, and related notes. The dashboard provides a focused view of medicines relevant to the current day and their medication status.

CareDose also supports OCR-assisted medicine or prescription data capture as an extension of manual entry, health-trend visualization using recorded values, and controlled sharing of selected health information. The emergency QR concept is designed to avoid unnecessary exposure of complete medical records by allowing only limited information to be displayed. The caretaker module is intended to support trusted assistance, including viewing permitted patient information and updating medication-taking status where authorized.

The frontend prototype is developed using HTML, CSS, and JavaScript. The backend is planned and developed using Python with Django REST Framework, while SQLite is used as the database for the academic prototype. The system follows a modular architecture so that authentication, profile management, medication management, reminders, health records, caretaker access, QR generation, and administration can be developed and maintained independently.

CareDose is intended as a support and organization tool rather than a replacement for professional medical advice. Its primary purpose is to improve medication-management consistency, record organization, accessibility of essential health information, and communication between a user and a trusted caretaker.

KEYWORDS: CareDose, Medication Management, Smart Medicine Reminder, Healthcare Support System, Medication Adherence, Prescription Management, Health Record Management, OCR-Based Data Extraction, Caretaker Access, Emergency QR Code, Health Monitoring, Django REST Framework.

Mapping with Sustainable Development Goals (Mention the Goal)	SDG 3 – Good Health and Well-Being SDG 9 – Industry, Innovation and Infrastructure SDG 10 – Reduced Inequalities

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ACKNOWLEDGEMENT	**v**
	ABSTRACTION	**vi**
	TABLE OF CONTENT	**vii**
	List of Figures	**xi**
	List of Tables	**xii**
	List of Abbreviations	**vii**
Chapter 1	INTRODUCTION	1
1.1	Background	1
1.2	Motivation	1
1.3	Scope & Objectives	1
1.4	Problem Statement	2
Chapter 2	REQUIREMENTS AND ANALYSIS	4
2.1	Existing System & Problem Statement	4
2.1.1	Limitations of Existing System	4
2.2	Solution Overview	4
2.3	Feasibility Study	5
2.3.1	Technical Feasibility	5
2.3.2	Economic Feasibility	6
2.3.3	Operational Feasibility	6
Chapter 3	LITERATURE REVIEW	9
3.1	PAPER 1 – Smart Medicine Reminder System Using OCR with Family Monitoring	9
3.2	PAPER 2 – Do Mobile Device Apps Designed to Support Medication Adherence Demonstrate Efficacy?	9
3.3	PAPER 3 – Mobile Apps for Increasing Treatment Adherence: Systematic Review	9
3.4	PAPER 4 – Mobile Phone Apps to Improve Medication Adherence	10

CHAPTER NO.	TITLE	PAGE NO.
3.5	PAPER 5 – The Effectiveness of Electronic Reminders to Improve Adherence	10
3.6	PAPER 6 – Mobile Telephone Text Messaging for Medication Adherence in Chronic Disease	10
3.7	PAPER 7 – Effect of Interactive eHealth Interventions on Medication Adherence	10
3.8	PAPER 8 – Electronic Multi-Compartment Medication Devices with Reminder Systems	10
3.9	PAPER 9 – Digital Solutions for Informal Caregivers and Medication Adherence	11
3.10	PAPER 10 – Image-Based Extraction of Prescription Information using OCR-Tesseract	11
3.11	Summary of the Literature Review	11
3.12	Research Gap Identified	11
Chapter 4	DIAGRAMS	13
4.1	Data Flow Diagrams (DFD)	13
4.1.1	DFD Level 0	13
4.1.2	DFD Level 1 – Patient/User	13
4.1.3	DFD Level 1 – Caretaker	14
4.2	Use Case Diagram	14
4.3	Class Diagram	15
4.4	State Diagram – Medication Status	16
4.5	Activity Diagram	17
4.6	Sequence Diagram	18
Chapter 5	METHODOLOGY	20
5.1	Module Descriptions	20
5.1.1	Authentication and Profile Module	20
5.1.2	Medicine and Prescription Management Module	21
5.1.3	Reminder and Medication Tracking Module	22
5.1.4	Health Record Management Module	23
5.1.5	Caretaker Module	23
5.1.6	Emergency Health QR Module	24
5.1.7	OCR Module	25

CHAPTER NO.	TITLE	PAGE NO.
5.1.8	Health Trend Dashboard	26
5.2	Algorithms	29
5.2.1	User Registration Algorithm	29
5.2.2	User Login Algorithm	29
5.2.3	Medicine Management Algorithm	29
5.2.4	Medication Status Algorithm	30
5.2.5	Caretaker Access Algorithm	30
5.2.6	OCR Processing Algorithm	30
5.2.7	Emergency QR Algorithm	30
5.3	Architecture	31
5.4	Case Study	33
Chapter 6	IMPLEMENTATION DETAILS	35
6.1	Implementation Procedures	35
6.1.1	Development Environment	35
6.2	Frontend Implementation	36
6.3	Backend Implementation	36
6.4	Database Implementation	37
6.5	Authentication and Profile	37
6.6	Medicine and Prescription Management	37
6.7	Reminder and Medication Tracking	38
6.8	Health Record Management	38
6.9	Caretaker Module	39
6.10	Emergency Health QR	39
6.11	OCR Support	40
6.12	Dashboard and Health Trends	40
6.13	Admin Module	40
6.14	Testing Methods and Results	41
Chapter 7	ADVANTAGES & DISADVANTAGES	43
7.1	Advantages	43
7.1.1	Centralized Medication Management	43
7.1.2	Personalized User Profile	43
7.1.3	Medication History	43
7.1.4	Caretaker Support	43

CHAPTER NO.	TITLE	PAGE NO.
7.1.5	Controlled Emergency Information	43
7.1.6	OCR-Assisted Entry	43
7.1.7	Health Trend Visualization	43
7.1.8	Modular Architecture	43
7.1.9	Account-Level Data Separation	43
7.1.10	Expandability	**Error! Bookmark not defined.**
7.2	Disadvantages	44
7.2.1	Internet/Browser Dependency	44
7.2.2	User-Provided Data	44
7.2.3	Privacy and Security Requirements	44
7.2.4	OCR Accuracy Limitations	44
7.2.5	Reminder Limitations in a Prototype	44
7.2.6	QR Privacy Risk	44
7.2.7	Not a Medical Diagnostic System	44
7.2.8	Technical Maintenance	44
Chapter 8	RESULT & DISCUSSION	45
8.1	System Output Overview	45
8.2	Authentication Results	45
8.3	Medicine Management Results	46
8.4	Reminder and Tracking Results	46
8.5	Caretaker Interaction Results	47
8.6	Health Record and Trend Results	47
8.7	OCR and QR Results	48
8.8	System Performance and Discussion	48
Chapter 9	CONCLUSION	50
9.1	Summary	50
9.2	Limitations	50
9.3	Future Scope	51
9.4	Applications	51
Chapter 10	APPENDICES	53

CHAPTER NO.	TITLE	PAGE NO.
Chapter 11	REFERENCES	59

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
4.1.1	DFD Level 0	13
4.1.2	DFD Level 1	14
4.1.3	DFD Level 1	14
4.2.1	Use Case Diagram	15
4.3.1	Class Diagram	16
4.4.1	Medication State Diagram	17
4.5.1	Activity Diagram	18
4.6.1	Sequence Diagram	19
6.2.1	Login Page	20
6.2.2	Dashboard	21
6.6.1	Medicine Management Interface	22
6.7.1	Reminder and Medication Tracking Module	22
6.8.1	Health Record Interface	23
6.9.1	Caretaker Interface	23
6.10.1	Emergency Health QR	24
6.11.1	OCR Feature	25
6.12.1	Health Trend Chart	27

LIST OF TABLES

TABLE NO.	TITLE	PAGE NO.
2.3	Hardware Requirements	7
2.4	Software Requirements	8
5.1	Major Modules	28
5.2	Main Database Entities	33
6.1	Development Technologies	36
6.3	Test Cases	42

LIST OF ABBREVIATIONS

ABBREVIATION		FULL FORM
AI	-	Artificial Intelligence
API	-	Application Programming Interface
CRUD	-	Create, Read, Update, Delete
CSS	-	Cascading Style Sheets
DB	-	Database
DFD	-	Data Flow Diagram
DRF	-	Django REST Framework
ER	-	Entity Relationship
HTML	-	HyperText Markup Language
HTTP	-	Hypertext Transfer Protocol
OCR	-	Optical Character Recognition
QR	-	Quick Response
REST	-	Representational State Transfer
SQL	-	Structured Query Language
UI	-	User Interface
URL	-	Uniform Resource Locator

CHAPTER 1

INTRODUCTION

1.1 Background

Medication management is an important part of everyday healthcare, particularly for people who need to take multiple medicines at different times, follow a fixed dosage, continue a medicine for a specific duration, or depend on another person for medication-related support. In practical situations, users may remember the medicine name but forget the exact time, food-related instruction, dose, or whether a dose has already been taken.

Traditional medication management often depends on handwritten prescriptions, medicine strips, paper notes, phone alarms, or memory. These methods can help with individual tasks but do not necessarily provide a unified record of medicines, medication status, health information, caretaker access, and historical records.

CareDose is proposed as a centralized web-based medicine reminder and health companion. It provides a structured environment where a user can maintain a personal profile, add medicines, define dosage and timing information, track whether doses are taken or pending, view medication history, maintain selected health records, and share controlled information with a caretaker.

The system also incorporates additional digital features such as OCR-assisted data capture, health-trend visualization, and an emergency health QR mechanism. These features are intended to reduce repetitive manual entry, improve accessibility of relevant information, and provide a compact way to access selected emergency details.

1.2 Motivation

The motivation for CareDose comes from the everyday difficulty of remembering medication schedules and maintaining health information in an organized manner. A medicine reminder alone may alert the user, but a useful health companion should also maintain what medicine was prescribed, how it should be taken, whether it was taken, and what relevant health information may need to be shared with a trusted person.

Another motivation is the need for controlled assistance. A caretaker may need to know whether medicines were taken or view permitted patient information, but unrestricted access to a complete medical profile can create privacy concerns. CareDose therefore treats caretaker access and emergency information sharing as controlled functions rather than unrestricted data exposure.

1.3 Scope & Objectives

The scope of CareDose covers a web-based system for medication organization and basic health-record support.

The main scope includes user authentication, profile management, medicine and prescription management, medication reminders and status tracking, medication history, health records, caretaker access, emergency QR-based information sharing, OCR-assisted medicine/prescription capture, health-trend visualization, and administrative management.

The current academic prototype uses a web architecture with HTML, CSS, and JavaScript on the frontend and Python/Django REST Framework with SQLite on the backend.

The architecture is designed so that the frontend can communicate with backend REST endpoints and the database can maintain structured records.

- To develop a centralized platform for storing and managing medicine information.
- To allow users to create profiles and maintain relevant personal and health information.
- To provide medication reminders and medication-status tracking.
- To maintain a history of taken and pending medication events.
- To allow trusted caretaker access to permitted patient information.
- To provide an emergency health QR facility with limited and controlled information exposure.
- To support OCR-assisted extraction of medicine or prescription information where applicable.
- To visualize selected health measurements and trends in a simple dashboard.
- To provide an administrative interface for controlled system management.
- To design the system in a modular way so that future notification services, integrations, and healthcare features can be added.

1.4 Problem Statement

People who take medicines regularly may find it difficult to remember medicine names, dosages, timings, frequency, duration, and specific instructions related to food intake.

They may also struggle to keep track of whether a scheduled dose has already been taken or missed. Medication-related information is often scattered across prescriptions, medicine packages, handwritten notes, phone alarms, and separate health applications, making it difficult to manage medicines systematically and review previous medication records.

Another important concern is the need for reliable assistance from family members or caretakers, particularly when patients require support in following their medication schedules. Caretakers may need access to medicine details, dose status, and relevant health information to help monitor the patient's routine.

In emergency situations, accessing essential health details quickly can also be important. However, personal health information is sensitive and must be protected through appropriate access controls to prevent unnecessary disclosure.

CareDose aims to address these challenges by integrating medicine management, scheduled reminders, dose-status tracking, personal health records, medication history, caretaker support, and controlled emergency information sharing within a single web-based application. By bringing these functions together, the system seeks to make medication management more organized, improve access to relevant health information, and support communication between patients and their caretakers while maintaining appropriate limits on information access.

Managing multiple medicines can be challenging due to different dosages, schedules, food instructions, and treatment durations. Patients may forget doses, struggle to maintain medication history, or find it difficult to monitor their health progress. Caretakers may also lack timely information about a patient's medication status, while emergency situations require quick access to essential health details. Therefore, an integrated system like CareDose can help simplify medication management, improve tracking, support caretaker involvement, and provide controlled access to important health information.

CHAPTER 2

REQUIREMENTS AND ANALYSIS

2.1 Existing System & Problem Statement

The existing approach to medication management commonly relies on paper prescriptions, medicine packaging, handwritten notes, calendar entries, basic phone alarms, or separate health applications. A user may use one method to remember the time and another method to store medical details. In many cases, medication history is maintained manually or not maintained at all.

Caretaker involvement is also commonly handled through direct communication, phone calls, or informal observation. This does not provide a structured record of medication status. Similarly, emergency medical information may be carried on paper or stored in a phone, which may not be immediately accessible in every situation.

2.1.1 Limitations of Existing System

Medication information may be scattered across prescriptions, notes, and medicine packages.

Simple alarms may remind a user of a time but do not maintain complete medicine records.

Manual tracking of taken and missed doses can be inconvenient.

Medication history may be difficult to retrieve when required.

Caretakers may not have a structured method for viewing permitted medication status.

Sharing complete medical information can create privacy concerns.

Emergency information may not be available in a compact, controlled format.

Repeated manual entry of medicine information can be time-consuming.

Separate applications may be required for medicine reminders, health records, and health-trend tracking.

2.2 Solution Overview

CareDose provides a unified web-based platform for medication management and health-support functions. After authentication, the user can access a dashboard containing relevant medicine information and health-related features. Medicines can be added with details such as name, dose, timing, frequency, food timing, duration, and notes. The system can maintain the medication state as Taken or Pending and preserve history for later review.

The user can maintain a profile containing relevant health information such as medical conditions, allergies, emergency contact information, weight, pulse, and other permitted records. A caretaker can be associated with the user and can access only the information made available through the caretaker

function. The system can also support OCR-assisted capture of medicine/prescription information and graphical display of selected health measurements.

The emergency health QR feature is intended to provide selected information quickly while avoiding the unnecessary exposure of a complete medical profile. The exact information included in the QR should therefore be limited to fields considered appropriate for emergency use.

2.3 Feasibility Study

A feasibility study is conducted to determine whether the proposed CareDose system can be developed and implemented successfully within the available resources, budget, and project duration. The study examines whether the system can provide a practical and affordable solution for managing medicines, scheduling reminders, recording dose status, maintaining personal health records, scanning prescriptions using OCR, visualizing health trends, supporting caretaker access, and sharing selected emergency information through a QR code. From an economic perspective, CareDose can be developed using open-source technologies such as Python, Django REST Framework, and SQLite, which helps reduce software development and licensing costs. The use of existing development tools and libraries also minimizes additional expenses. From a schedule perspective, the project can be completed within the available academic period by implementing the essential features first and improving additional functionalities as time permits. Overall, the feasibility study indicates that CareDose is a practical and cost-conscious academic project that can be developed using available resources while addressing common medication management challenges.

2.3.1 Technical Feasibility

The project is technically feasible because the required development technologies are widely available. H The CareDose project is technically feasible because it uses widely available technologies and development tools that support web application development. HTML, CSS, and JavaScript are used to develop the frontend interface, while Python and Django REST Framework (DRF) provide a structured backend for authentication, data processing, business logic, and API communication. SQLite is selected as the database management system for the academic prototype because it is lightweight, easy to configure, and does not require a separate database server. The use of Django ORM simplifies database operations and supports organized data management. Supporting technologies, including OCR processing, QR code generation, graphical visualization, and the Twilio API for SMS reminders, extend the application's functionality. The modular architecture separates the presentation, application, and data layers, making the system easier to maintain and enhance. If

the application is expanded to support more users and greater concurrent access, the database can be migrated to a server-based relational database such as PostgreSQL, with the necessary configuration and testing. Therefore, the technologies selected provide a practical foundation for developing and evaluating CareDose while allowing scope for future improvements.

TML, CSS, and JavaScript provide the frontend layer, while Python and Django REST Framework provide a structured backend API. SQLite is suitable for an academic prototype because it is lightweight and does not require a separate database server. The modular architecture also permits future migration to a larger database if required.

2.3.2 Economic Feasibility

The CareDose project is economically feasible for academic development because most of the required programming languages, frameworks, and development tools are available free of charge or as open-source software. Technologies such as HTML, CSS, JavaScript, Python, Django REST Framework, and SQLite do not require expensive software licences for developing the prototype. The project can be developed and tested using an existing computer or laptop, a standard web browser, and internet connectivity. Visual Studio Code also provides a suitable development environment without requiring a paid licence. However, certain additional services may involve costs depending on usage and deployment requirements. These may include web hosting, cloud storage, third-party API services, and SMS notifications through Twilio. Production deployment may also require additional database infrastructure and security measures. Since the current project is intended as an academic prototype, the overall development cost can be kept relatively low by using locally available resources and free development tools, while future operational expenses can be considered when expanding the system.

2.3.3 Operational Feasibility

The CareDose project is operationally feasible because its main features are organized around familiar medication-management activities and can be accessed through a user-friendly web interface. Patients can manage medicine records, view medication schedules, update dose statuses, maintain personal health records, and review health trends through graphical representations. The system also supports SMS medication reminders through the Twilio API and provides OCR-assisted prescription entry and Emergency Health QR functionality. Trusted caretakers can access permitted information about linked patients and provide assistance with medication monitoring, depending on the implemented permissions. The interface is designed to be understandable for first-time users, with minimal training

required to perform common operations. Since the application uses commonly available web technologies, it can be accessed through a compatible browser without requiring a dedicated mobile application. CareDose is intended to complement existing healthcare routines by supporting medication organization and personal health record keeping. It does not replace professional medical advice, doctors, pharmacists, or emergency medical services.

Sl. No.	**Hardware Component**	**Minimum Requirement**
1	Processor	Intel Core i3 or equivalent
2	RAM	4 GB minimum; 8 GB recommended
3	Storage	20 GB free disk space
4	Display	1366 × 768 resolution or higher
5	Network	Stable internet connection

2.3 Hardware Requirement

Sl. No.	Software/Technology	Purpose
1	Operating System	Provides the environment for development and execution
2	HTML	Structures the frontend pages
3	CSS	Styles the user interface
4	JavaScript	Provides frontend interactivity
5	Python	Supports backend development
6	Django REST Framework	Provides backend APIs
7	SQLite	Stores application data
8	Visual Studio Code	Development and code editing
9	Git and GitHub	Version control and code management
10	Chrome or Edge	Runs and tests the web application
11	Twilio	The purpose of Twilio in CareDose is to send SMS medicine reminders to patients' mobile

2.4 Software Requirements

CHAPTER 3

LITERATURE REVIEW

The literature review examines research and digital-health approaches related to medication adherence, medicine reminders, OCR-based prescription capture, interactive eHealth, and caregiver-supported medication management. The following studies provide the technical and functional background relevant to the development of CareDose.

3.1 PAPER 1 – Smart Medicine Reminder System Using OCR with Family Monitoring

The base paper selected for CareDose is "Smart Medicine Reminder System Using OCR with Family Monitoring," published in the International Journal of Engineering Development and Research (IJEDR), Volume 14, Issue 2, April 2026. The work is directly related to the project because it combines medicine-reminder functionality with OCR-based information capture and family monitoring. These concepts support the CareDose design of medicine scheduling, OCR-assisted entry, and controlled caretaker involvement.

3.2 PAPER 2 – Do Mobile Device Apps Designed to Support Medication Adherence Demonstrate Efficacy?

Armitage, Kassavou and Sutton presented a systematic review and meta-analysis of randomized controlled trials evaluating mobile-device applications for medication adherence. The review examined nine trials involving 1,159 recruited participants and reported an association between app-based interventions and higher self-reported medication adherence, while also noting limitations such as small samples and reliance on self-reported measures. The findings support the inclusion of reminders and adherence tracking in digital medication-support systems.

3.3 PAPER 3 – Mobile Apps for Increasing Treatment Adherence: Systematic Review

Pérez-Jover, Sala-González, Guilabert and Mira reviewed studies on mobile applications for treatment adherence. The review found that several included studies reported improvements in adherence and that users generally considered the applications useful for medication management. The study supports the CareDose approach of providing an organized digital interface for medication schedules, reminders and medication history.

3.4 PAPER 4 – Mobile Phone Apps to Improve Medication Adherence

Santo and colleagues conducted a systematic assessment of mobile phone applications intended to improve medication adherence. The study examined application features such as flexible scheduling, medication-history tracking, snooze functions and visual aids. This work is relevant to CareDose because the project similarly treats scheduling, history and user interaction as important components rather than providing only a basic alarm.

3.5 PAPER 5 – The Effectiveness of Electronic Reminders to Improve Adherence

Vervloet and colleagues reviewed electronic reminder interventions for chronic medication adherence. The review considered SMS reminders, electronic reminder devices and pager-based interventions and found evidence of short-term effectiveness in several studies, while noting uncertainty about longer-term effects. This supports the CareDose emphasis on timely reminders while also motivating the project's acknowledgement that reliable long-term notification services require appropriate deployment infrastructure.

3.6 PAPER 6 – Mobile Telephone Text Messaging for Medication Adherence in Chronic Disease

A meta-analysis of randomized clinical trials examined mobile telephone text messaging as a method for supporting medication adherence. The study reported improved adherence in the pooled analysis but also highlighted variation in study duration, disease populations and adherence measurement. The work indicates that reminder systems are more useful when combined with mechanisms for recording and reviewing medication-taking behaviour, which is reflected in CareDose's tracking and history features.

3.7 PAPER 7 – Effect of Interactive eHealth Interventions on Medication Adherence

Pouls and colleagues reviewed interactive eHealth interventions for adults using long-term medication. The review included interventions aimed at patients or caregivers and identified evidence supporting approaches such as SMS, mobile applications, calls, medication-management education and communication support. The findings are relevant to CareDose's combination of patient functions and controlled caretaker access.

3.8 PAPER 8 – Electronic Multi-Compartment Medication Devices with Reminder Systems

Paterson, Kinnear, Bond and McKinstry reviewed electronic multi-compartment medication devices that use audible or visual reminders and, in some cases, contact caregivers. The review highlighted the importance of usability and context when evaluating reminder systems. CareDose applies the same general principle in software form by linking medication schedules with status tracking and optional caretaker support.

3.9 PAPER 9 – Digital Solutions for Informal Caregivers and Medication Adherence

Espírito-Santo, Santos and Estêvão examined digital solutions available to informal caregivers for supporting medication adherence. The review identified tools including mobile applications, SMS-based systems and wearable technologies and emphasized the role of caregivers in medication management. This supports the inclusion of a dedicated caretaker role in CareDose with access controlled according to the intended patient-support relationship.

3.10 PAPER 10 – Image-Based Extraction of Prescription Information using OCR-Tesseract

The 2024 study "Image-Based Extraction of Prescription Information using OCR-Tesseract" examines image processing and OCR methods for extracting prescription information such as medicine names, dosages and instructions. The work demonstrates the relevance of preprocessing and OCR for reducing manual data entry. CareDose adopts OCR as an assistance feature, while requiring the extracted information to be reviewed before it is stored as a medicine record.

3.11 Summary of the Literature Review

The reviewed studies show that digital medication-support systems commonly use reminders, adherence tracking, mobile applications, electronic prompts, OCR, and caregiver support. The literature also indicates that usability, privacy, reliable notification delivery and appropriate user verification are important when healthcare-support applications are designed. These observations are reflected in the functional design of CareDose.

3.12 Research Gap Identified

Many existing approaches to medication management focus on individual aspects of patient care, such as medicine reminders, mobile-based medication adherence applications, Optical Character Recognition (OCR) for prescription scanning, or caregiver assistance. Although these approaches help patients manage their medication routines, they may not provide a complete solution that integrates medication management with personal health information, medication history, emergency support, and family monitoring. CareDose aims to address this limitation by combining these essential functions within a single web-based application.

The system integrates medicine scheduling and reminders, dose-status tracking, prescription scanning using OCR, personal health record management, medication history, and health-trend visualization to support patients in monitoring their medication routines and reviewing their health-related information. In addition, CareDose provides controlled access to selected emergency information through an Emergency Health QR feature, enabling important details to be accessed when required while limiting the information exposed. The application also incorporates patient, caretaker, and

administrative roles to support medication management, family assistance, and system-level administration.

Therefore, the main contribution of CareDose lies in the integration of multiple medication-support and health-information management functions within a unified academic web application. Rather than treating medication reminders as an isolated feature, the project emphasizes coordinated functionality, improved accessibility, caregiver involvement, and controlled information access. This integrated approach is intended to make medication management more organized and convenient while providing a foundation for further improvements in patient monitoring and healthcare support.

CHAPTER 4

DIAGRAMS

4.1 Data Flow Diagrams (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data move through a system. It shows the external entities that supply or receive data, the processes that transform the data, and the data stores in which data are kept. A DFD can be drawn at different levels of detail. The context-level (Level 0) diagram shows the system as a single process, and the Level 1 diagrams break that process into its main sub-processes. The DFDs of Talent Flow use request and response arrows between the entities, processes and data stores**.**

4.1.1 DFD Level 0

Figure 4.1.2 shows the Level 0 Data Flow Diagram (DFD) of the CareDose platform. The system is represented as a single process, CareDose, interacting with two external entities: the Patient/User and the Caretaker. The Patient/User provides information such as medicine details, prescriptions, and health records to the system and receives medication reminders, tracking information, health trends, and emergency information. The Caretaker provides and receives relevant medication and health-related information to support the user.

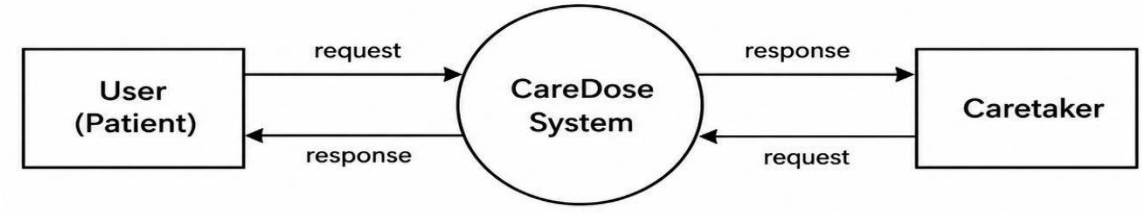


Figure 4.1.1 DFD Level 0

4.1.2 DFD Level 1 – Patient/User

Figure shows the Level 1 DFD of the CareDose system for the Patient, covering login, profile and health management, medicine management, reminders, health records, analytics,OCR. Each process interacts with the relevant data stores to securely manage and retrieve the patient's information**.**

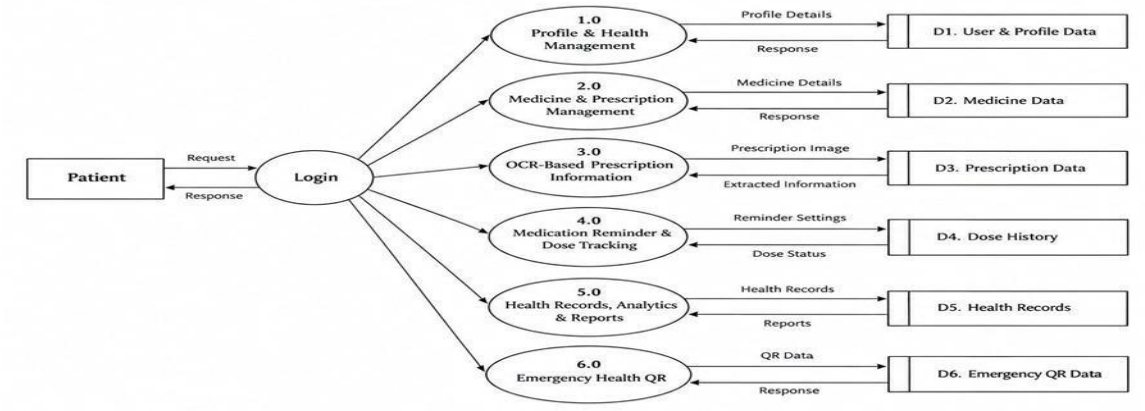


Figure 4.1.2 DFD Level 1

4.1.3 DFD Level 1 – Caretaker

Figure shows the Level 1 DFD for the Caretaker module of CareDose. It illustrates how the caretaker interacts with the system to monitor medication-related information and receive relevant alerts for the patient.

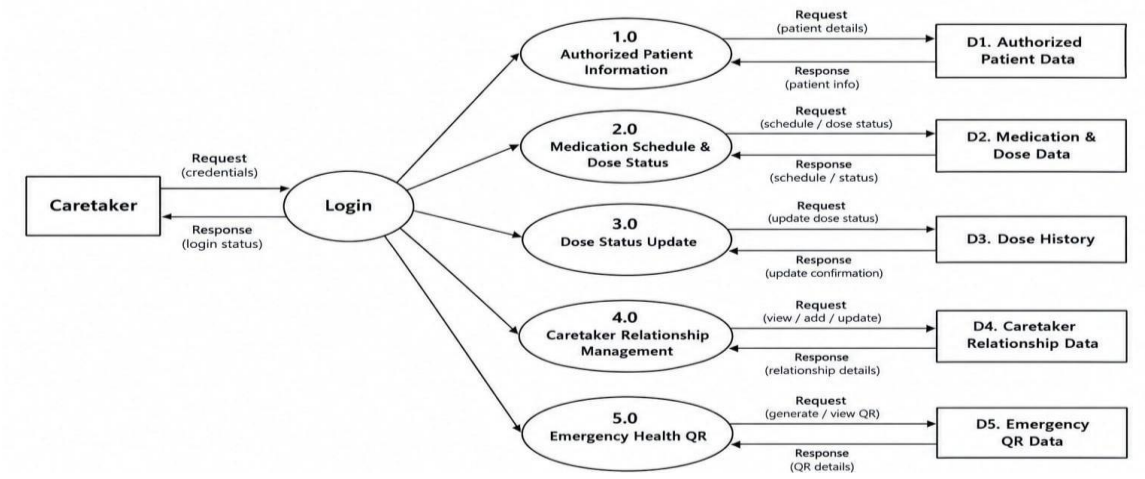


Figure 4.1.3 DFD Level 1

4.2 Use Case Diagram

Actors: User/Patient, Caretaker, Admin. Include register, login, profile, medicine CRUD, reminders, history, health records, QR, OCR, trends, caretaker access, admin functions.The User can register, log in, manage a profile, add and manage medicines, track medication status, view history, maintain health records, use OCR-assisted capture, view health trends, generate emergency QR information, and manage caretaker access. The Caretaker can log in through the permitted mechanism, view authorized information, and perform medication-status actions where enabled. The Admin manages the system-level administrative functions.

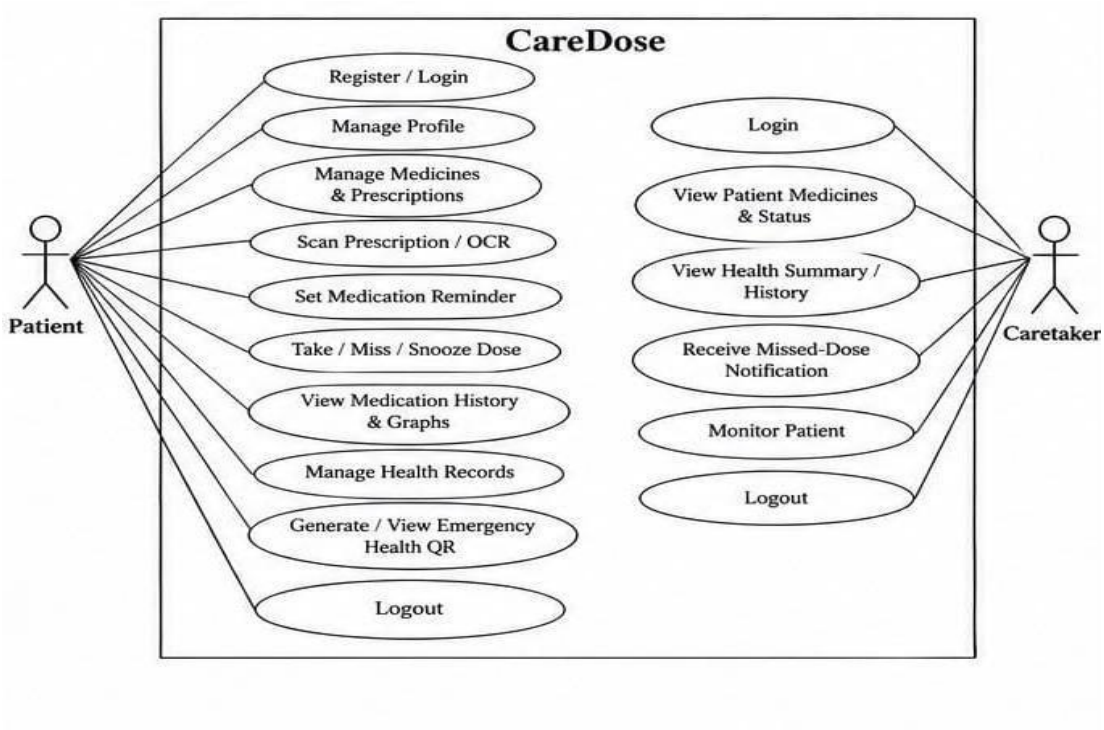


Figure 4.2.1 Use Case Diagram

4.3 Class Diagram

User, Profile, Medicine, Medication Schedule, Medication History, Health Record, Caretaker, QR Access, OCR Record, Admin. The class diagram should show the attributes and major methods of each class and the relationships between them. A User may have one Profile and multiple Medicine, Health Record, and Medication History records. A User may associate one or more caretaker records depending on the final implementation. Medicine records contain scheduling and dosage information.

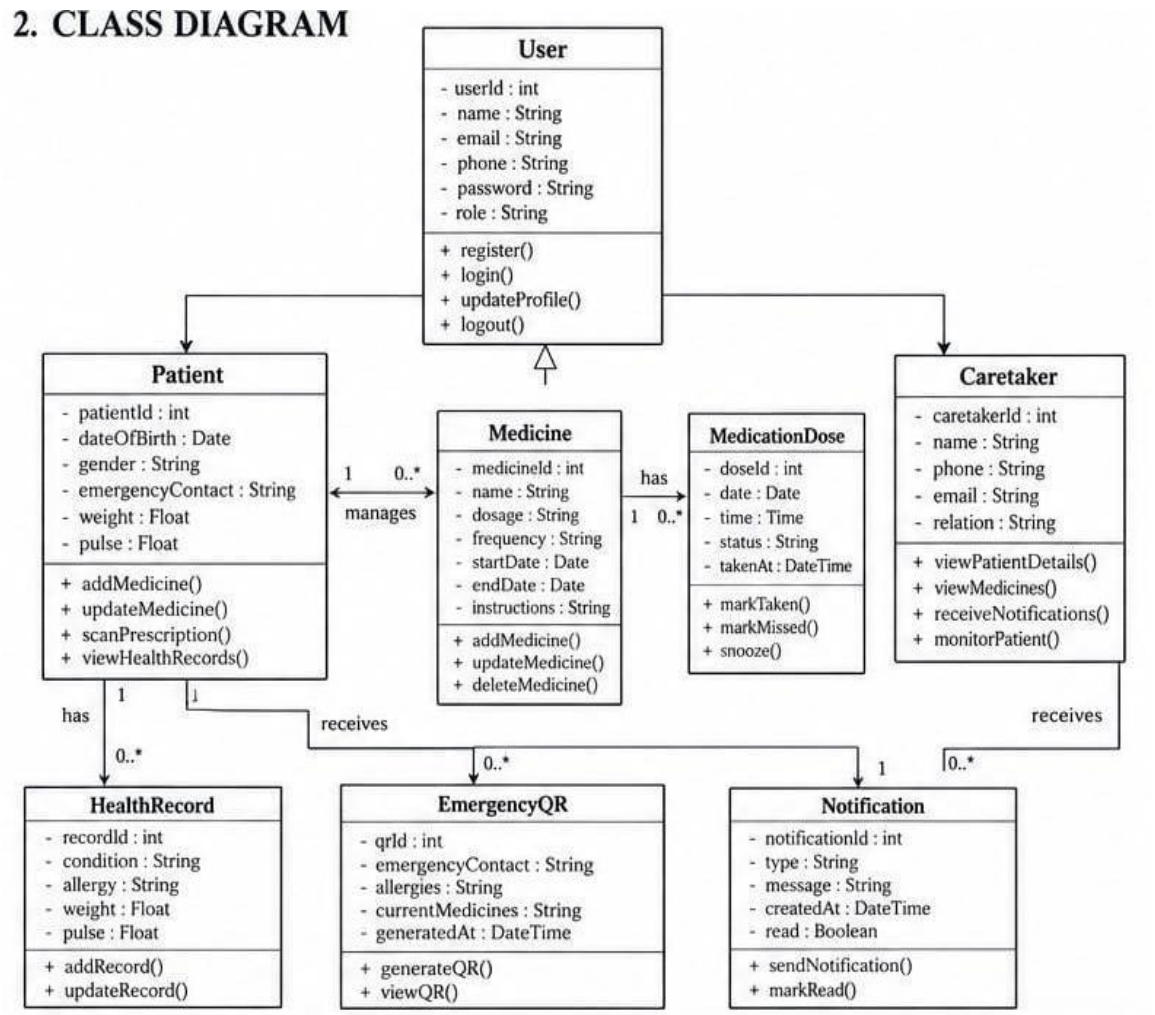


Figure 4.3.1 Class Diagram

4.4 State Diagram – Medication Status

Suggested flow: Scheduled/Pending → Taken; Scheduled/Pending → Missed; Pending → Snoozed → Pending; completed doses recorded in history.The medication state diagram represents the lifecycle of a scheduled dose. A dose can begin as Pending/Scheduled, move to Taken when the user or authorized caretaker records completion, or move to Missed when the defined reminder window expires. If snooze functionality is enabled, the reminder may temporarily enter a Snoozed state and return to Pending.

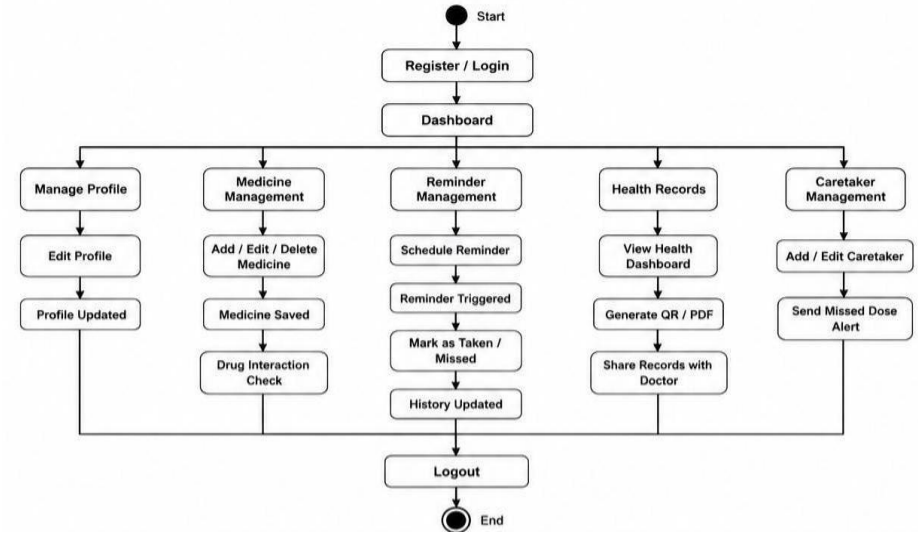


Figure 4.4.1 Medication State Diagram

4.5 Activity Diagram

The CareDose activity diagram represents the sequence of activities performed by users while interacting with the system. The process begins when the user opens the application and enters their login credentials. After successful authentication, the system redirects the user to the appropriate dashboard based on their role, such as patient, caretaker, or administrator. The patient can manage medicine records, view scheduled doses, update medication status, maintain health records, view graphical representations of health trends, scan prescriptions using OCR, and generate an Emergency Health QR code. The system stores relevant information in the SQLite database and uses Twilio to send SMS reminders for scheduled medicines. The caretaker can access permitted information about the linked patient and assist with medication monitoring, while the administrator can perform authorized system-level operations. If invalid login credentials are entered, the system displays an error message and requests the user to try again. The activity diagram illustrates these workflows and decision points, providing a clear representation of the overall operation of the CareDose application.

Show login → dashboard → select feature → perform operation → validate → store/retrieve → display result → logout.

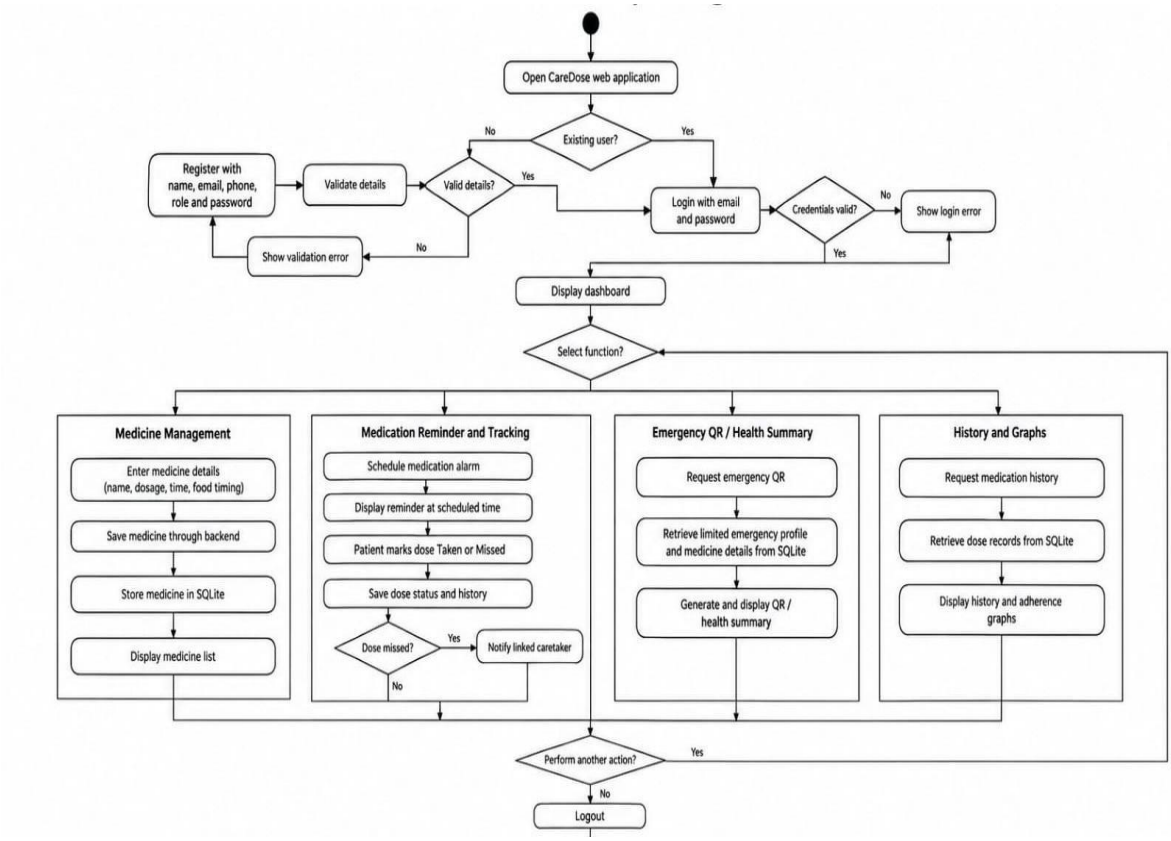


Figure 4.5.1 Activity Diagram

4.6 Sequence Diagram

Show User → Frontend → Django REST API → Database → API → Frontend for a medicine-add or medication-status operation.

The CareDose sequence diagram illustrates the sequence of interactions between the user, web interface, Django REST API, SQLite database, OCR component, and Twilio SMS service. The process begins when the patient, caretaker, or administrator accesses the web interface and submits login credentials. The frontend sends the authentication request to the Django REST API, which validates the credentials and retrieves the required user information from the database. After successful authentication, the appropriate dashboard is displayed according to the user's role. When the patient manages medicine records or updates medication status, the request is processed by the backend and the relevant information is stored in or retrieved from SQLite. For prescription scanning, the user uploads an image, the OCR component extracts readable medicine details, and the extracted information is presented for verification before the medicine record is saved. The system can also generate an Emergency Health QR code containing selected health information and display graphical representations of recorded health trends. For medication reminders, the application communicates

with the Twilio API to send SMS notifications to the configured phone number. The sequence diagram demonstrates how these components communicate to provide authentication, medicine management, medication tracking, health-record visualization, prescription scanning, emergency information access, and SMS reminder functionality.

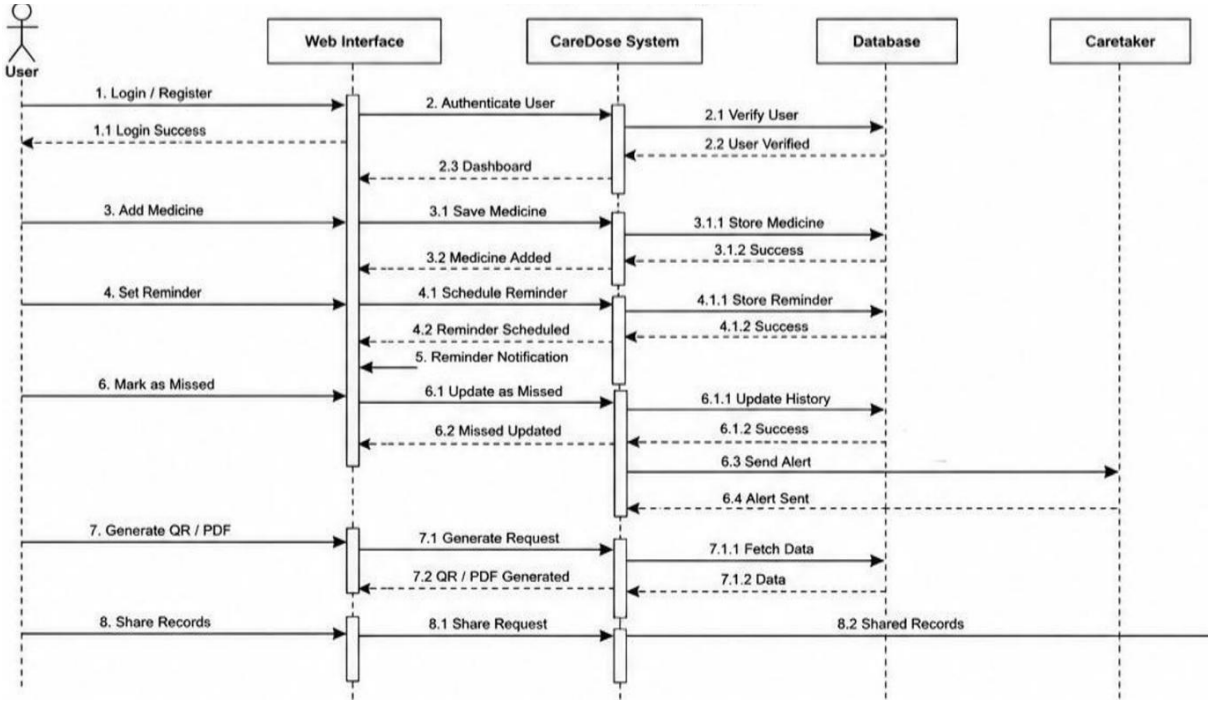


Figure 4.6.1 Sequence Diagram

CHAPTER 5

METHODOLOGY

5.1 Module Descriptions

5.1.1 Authentication and Profile Module

The Authentication and Profile module provides registration, login, logout, and profile management. The registration process validates required fields before creating an account. The profile stores permitted personal and health-related details such as medical conditions, allergies, emergency contact, weight, and pulse. The profile information is linked to the authenticated user.

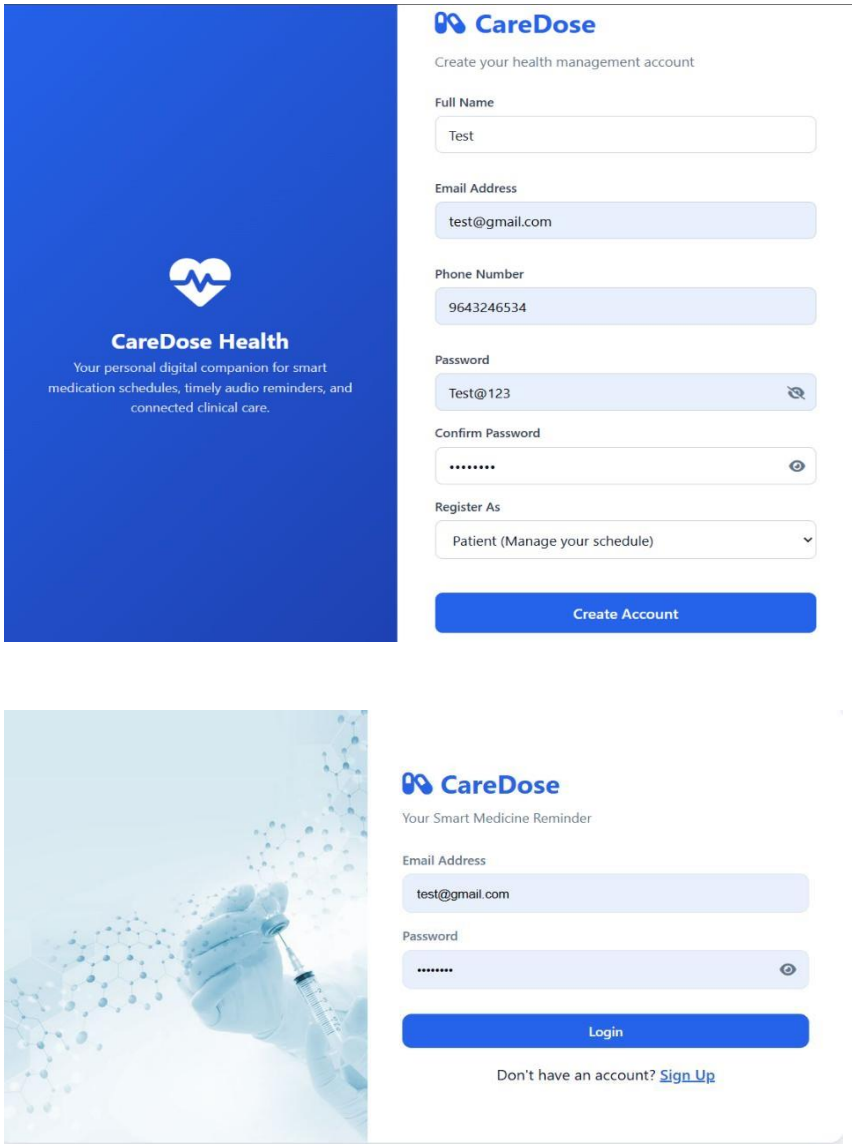


Figure 6.2.1 Login Page

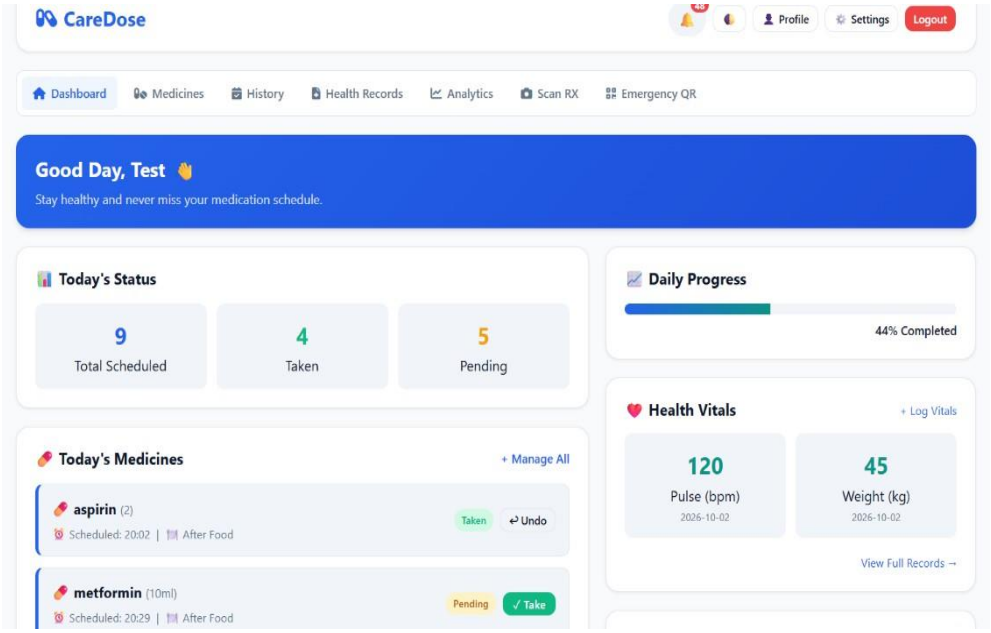


Figure 6.2.2 Dashboard

5.1.2 Medicine and Prescription Management Module

This module enables users to manage their medication records by adding new medicines, viewing existing details, editing information, and deleting records when necessary. Each medicine record includes essential details such as the medicine name, dosage, scheduled timing, frequency, treatment duration, food instructions, and additional notes. The module stores and retrieves medicine information through the backend and database, ensuring that records remain organized and available for future reference. It also supports medication scheduling and tracking by providing the information required for reminders and dose-status updates. Each medicine record is associated with the authenticated patient, helping maintain data separation and preventing one user's medication information from appearing in another user's account.

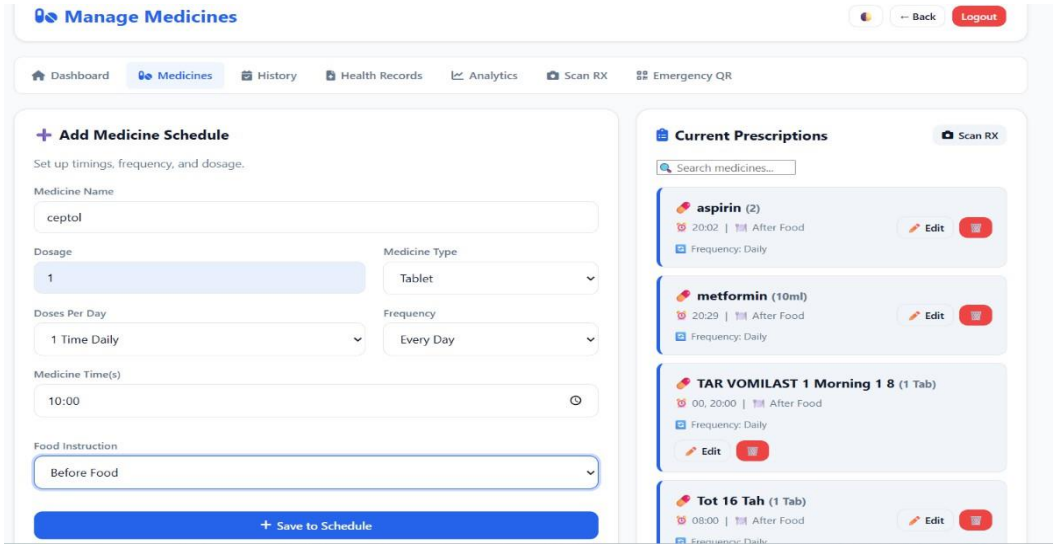


Figure 6.6.1 Medicine Management Interface

5.1.3 Reminder and Medication Tracking Module

The Reminder and Tracking Module helps users manage their medicines according to their prescribed schedules. The dashboard displays relevant medicines based on their scheduled timings, recurring schedules, and defined treatment periods or start dates. Users can view upcoming doses and update the status of each scheduled dose as Taken or Pending. The module supports medication adherence by helping users keep track of their daily medicine routine and identify doses that still require attention. It also maintains dose-status history, allowing users to review previously recorded medication activity and monitor their adherence over time. In addition, the scheduler can trigger SMS reminders through Twilio when the configured reminder time is reached and the scheduler is running, subject to messaging service availability.

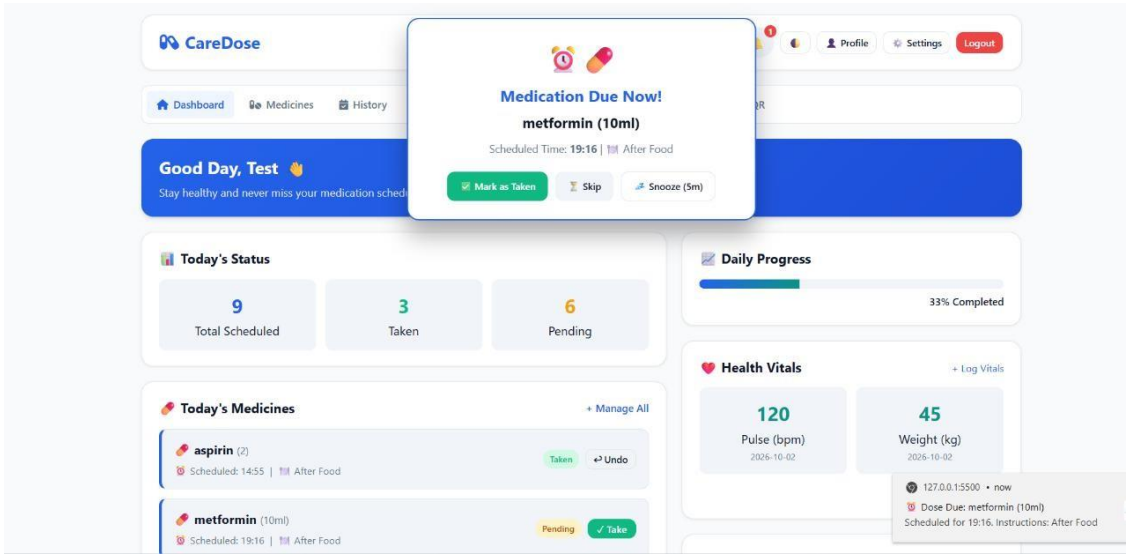


Figure 6.7.1 Reminder and Medication Tracking Module

5.1.4 Health Record Management Module

The Health Record Management module stores and organizes selected health information entered by the user, such as weight and pulse rate. Users can add and update their health records over time, allowing the system to maintain a history of recorded values. The module presents this information through simple graphical representations, such as charts, making it easier to observe changes and identify general trends. These records help users keep track of their health information and review previous measurements whenever needed. The module is designed for health record management and trend observation only and does not provide medical diagnosis or replace professional medical advice.

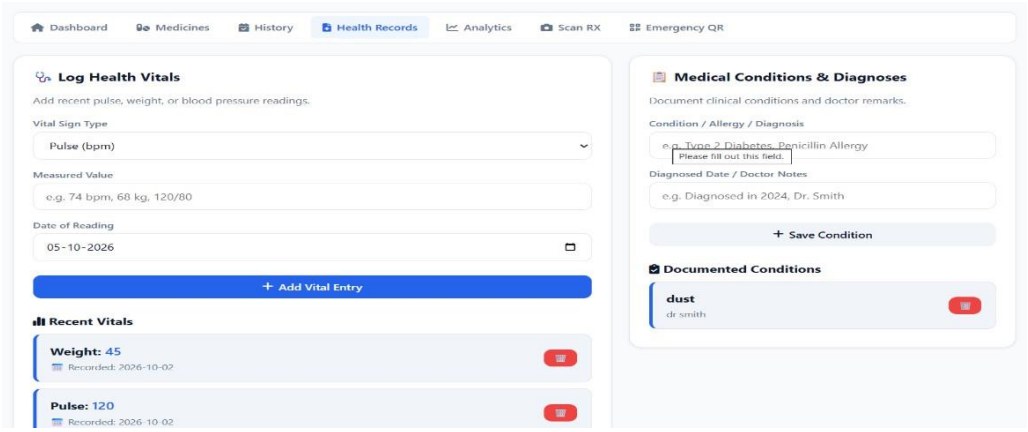


Figure 6.8.1 Health Record Interface

5.1.5 Caretaker Module

The Caretaker module is designed to help family members or trusted individuals support patients in managing their medication schedules and health information. The patient can establish a connection with a trusted caretaker, allowing the caretaker to access permitted patient details through the system. Depending on the implemented functionality, the caretaker can monitor medicine schedules, review medication status, and identify doses marked as Taken or Not Taken. This module helps improve communication between patients and their caretakers and supports better medication monitoring, particularly for elderly individuals or those who require assistance with their daily medication routine. Access is restricted to the information and functions permitted for the linked patient-care relationship, ensuring that the caretaker does not receive unrestricted administrative privileges.

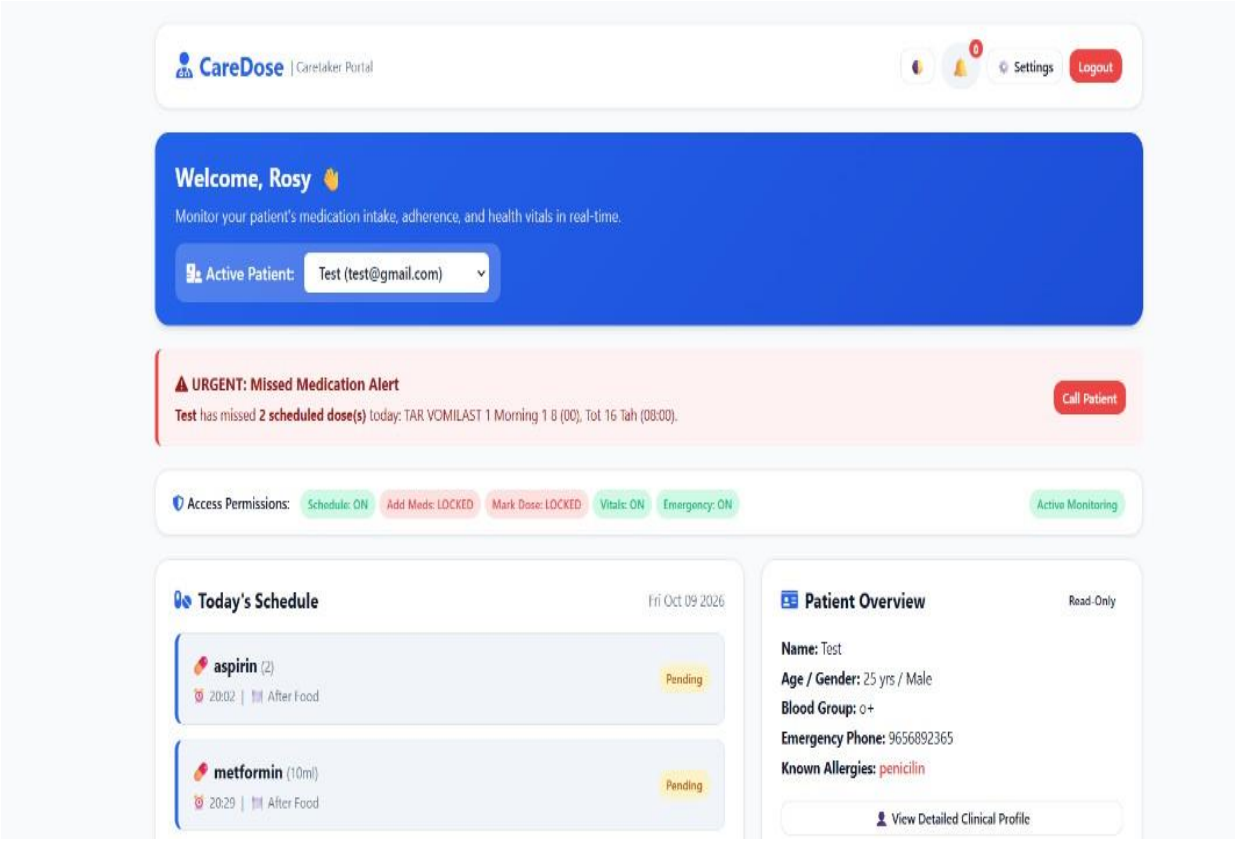


Figure 6.9.1 Caretaker Interface

5.1.6 Emergency Health QR Module

The Emergency Health QR module generates a QR code containing selected health information that can be accessed when quick access to essential details is required during an emergency. The QR code provides a convenient way to view important information such as the patient's name, known allergies, major medical conditions, emergency contact number, and other approved health details. This feature can be particularly useful when the patient is unable to communicate their medical information independently. By making essential details accessible through a QR code, the module aims to support faster access to relevant emergency information. The information displayed is intended to be limited to essential, user-approved details rather than the patient's complete health records. Future enhancements may include authenticated access, time-limited QR validity, and additional security mechanisms to improve privacy and prevent unauthorized access to sensitive health information.

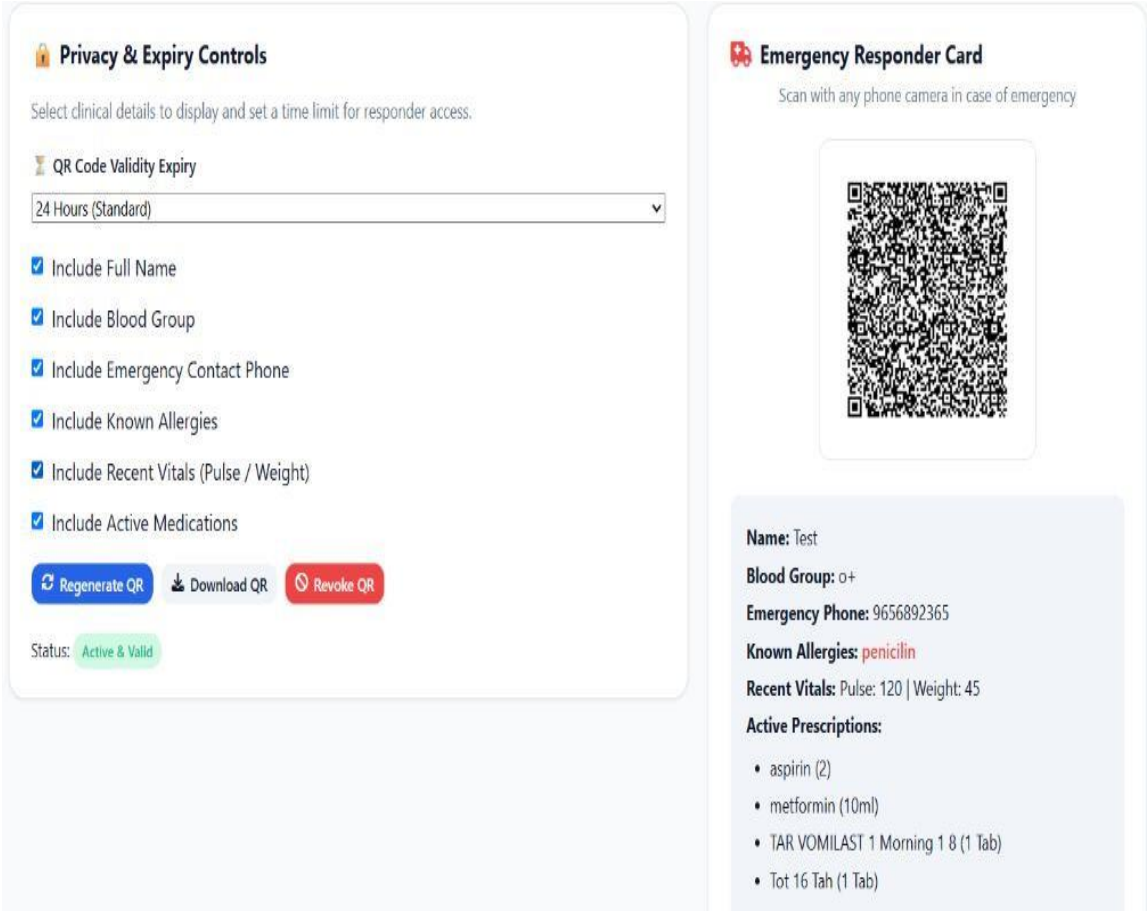


Figure 6.10.1 Emergency Health QR

5.1.7 OCR Module

The OCR (Optical Character Recognition) module is designed to extract readable text from uploaded medicine labels or prescription images, including medicine names, dosage details, and medication schedules where available. The extracted text helps reduce manual data entry and makes it easier for users to enter prescription information into the system. After processing the uploaded image, the system presents the recognized information for review before the final medicine record is saved. Since the accuracy of text extraction may vary depending on image quality, handwriting, and prescription clarity, the user must verify and correct the extracted details before saving them. This module aims to simplify medicine entry while maintaining user control over the accuracy of medication records.

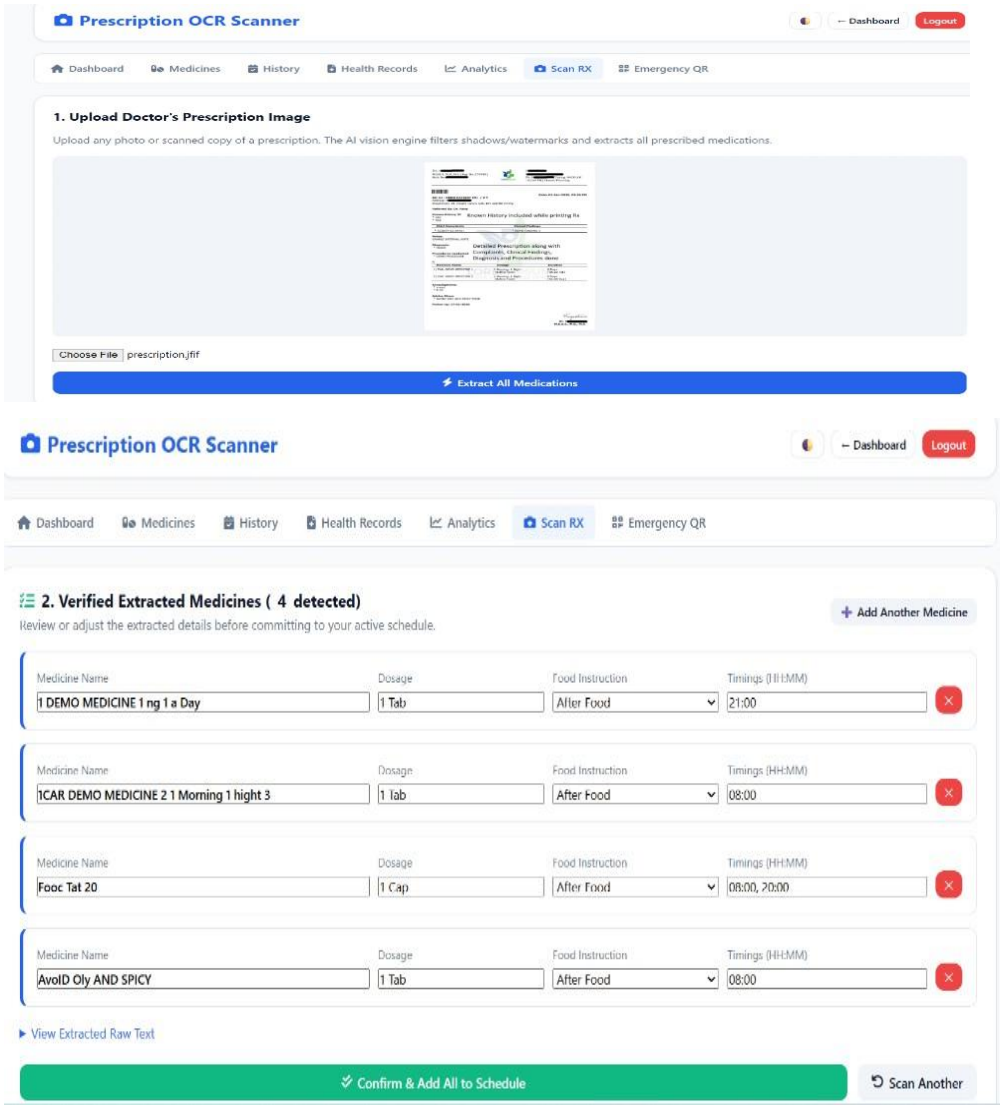


Figure 6.11.1 OCR Feature

5.1.8 Health Trend Dashboard

The Health Trends and Graphs module presents selected health records in a visual format to help users understand and review their recorded health information over time. The dashboard can display graphical representations of values such as weight and pulse rate plotted against their respective recorded dates. These charts allow users to observe changes, compare previous measurements, and identify general trends in their health records. By presenting information visually, the module makes health data easier to interpret than reviewing individual entries separately. The graphs are generated using the information entered and maintained by the user and are intended for personal monitoring and record review rather than medical diagnosis or treatment recommendations.

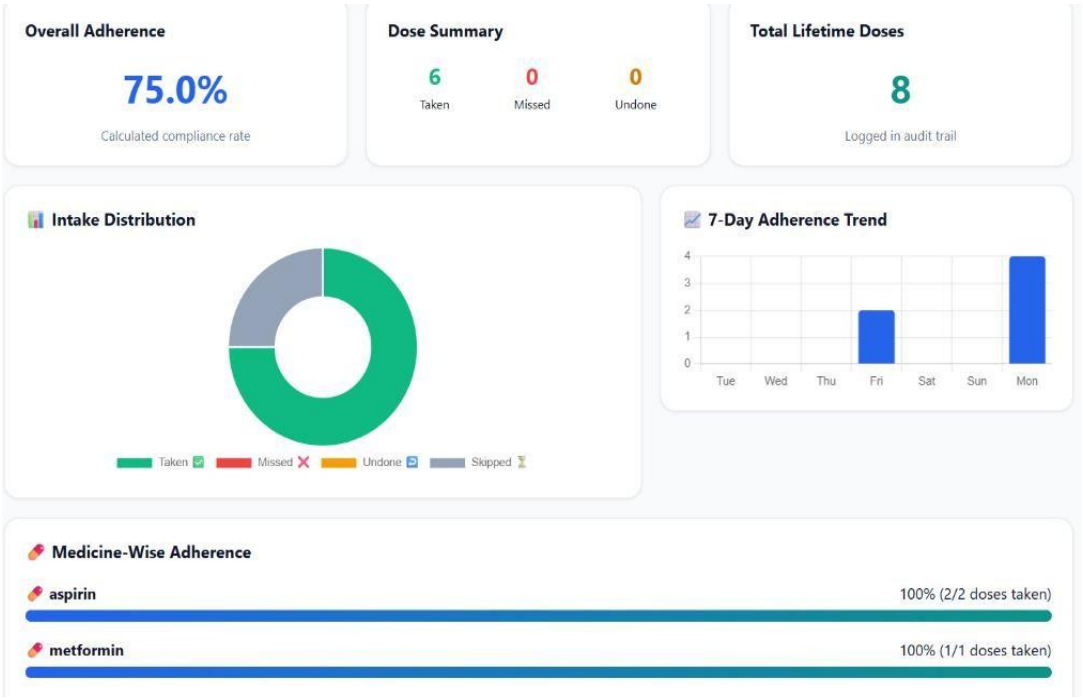


Figure 6.12.1 Health Trend Chart

Sl. No.	Module	Description
1	Authentication and Profile	Registration, login, logout and profile management; profile details are linked to the authenticated user.
2	Medicine and Prescription Management	Add, view, edit and delete medicine records with dosage, timing, frequency, duration and food instructions.
3	Reminder and Medication Tracking	Shows scheduled medicines, records dose status as Taken or Pending, keeps dose history and supports SMS reminders through Twilio.
4	Health Record Management	Stores selected health values such as weight and pulse rate and maintains a history of recorded values.
5	Caretaker	Lets a linked caretaker view permitted patient information and monitor medication status.
6	Emergency Health QR	Generates a QR code containing selected, user-approved emergency health details.
7	OCR	Extracts text from uploaded medicine labels or prescription images for the user to verify before saving.
8	Health Trend Dashboard	Displays graphs of recorded health values such as weight and pulse rate against their recorded dates.

5.1 Major Modules

5.2 Algorithms

5.2.1 User Registration Algorithm

1. Start.
2. Collect registration details.
3. Validate required fields and input format.
4. Check whether the account already exists.
5. If validation fails, display an appropriate error.
6. If valid, create the user record.
7. Create or initialize the associated profile.
8. Display successful registration and allow login.
9. End.

5.2.2 User Login Algorithm

1. Start.
2. Accept username/email and password.
3. Validate the input.
4. Search for the corresponding account.
5. Verify the credentials.
6. If invalid, display an error message.
7. If valid, create the authenticated session/token and redirect to the correct dashboard.
8. End.

5.2.3 Medicine Management Algorithm

1. Start.
2. Identify the authenticated user.
3. Accept medicine details.
4. Validate medicine name, dosage, schedule, and duration fields.
5. Create or update the medicine record.
6. Associate the medicine with the authenticated user.
7. Return the updated medicine list.
8. End.

5.2.4 Medication Status Algorithm

1. Start.
2. Identify today's scheduled medicine dose.
3. Display the dose as Pending/Scheduled.
4. When the user or authorized caretaker selects Taken, record the time and status.
5. If the dose is not completed within the defined reminder window, mark it as Missed where that rule is implemented.
6. Store the event in medication history.
7. Update the dashboard.
8. End.

5.2.5 Caretaker Access Algorithm

1. Start.
2. Authenticate the caretaker.
3. Identify the patient-caretaker relationship.
4. Check the permissions granted by the patient/system.
5. If access is not permitted, deny the request.
6. If permitted, retrieve only authorized patient data.
7. Allow permitted medication-status actions.
8. Record relevant changes.
9. End.

5.2.6 OCR Processing Algorithm

1. Start.
2. Accept a prescription or medicine image.
3. Preprocess the image where required.
4. Perform OCR text extraction.
5. Display the extracted text to the user.
6. Allow the user to correct or confirm medicine information.
7. Save the verified information as a medicine record.
8. End.

5.2.7 Emergency QR Algorithm

1. Start.
2. Authenticate the user.

3. Collect only the fields selected for emergency sharing.
4. Create the QR representation.
5. Display the QR to the user.
6. When scanned, display only the permitted information according to the implemented access rule.
7. End.

5.3 Architecture

CareDose follows a layered web architecture that separates the user interface, application logic, and data storage to support organized development and maintenance.

The presentation layer is developed using HTML, CSS, and JavaScript and provides the interfaces through which patients, caretakers, and administrators interact with the system. The application layer is implemented using Python and Django REST Framework (DRF), which handles authentication, input validation, business logic, API endpoints, and communication between the frontend and the database. SQLite is used as the database management system to store and retrieve persistent information, including user profiles, medicine records, medication schedules, dose statuses, and health records.

The general data flow follows the sequence: User/Caretaker/Admin → CareDose Web Interface → Django REST API → SQLite Database.

When a user performs an operation, the frontend sends the relevant request to the backend API. The backend processes the request, applies the necessary validation and business rules, and communicates with the database when data needs to be stored or retrieved.

The processed response is then returned through the API and displayed on the user interface. Supporting components are integrated into the appropriate layers: OCR processes uploaded prescription images to extract medicine details, QR generation provides access to selected emergency health information, and chart-rendering functionality presents recorded health data visually.

Twilio is integrated through the application layer to support SMS medication reminders. This layered architecture helps maintain a clear separation of responsibilities and supports the implementation of CareDose's core healthcare-support functionalities.

.

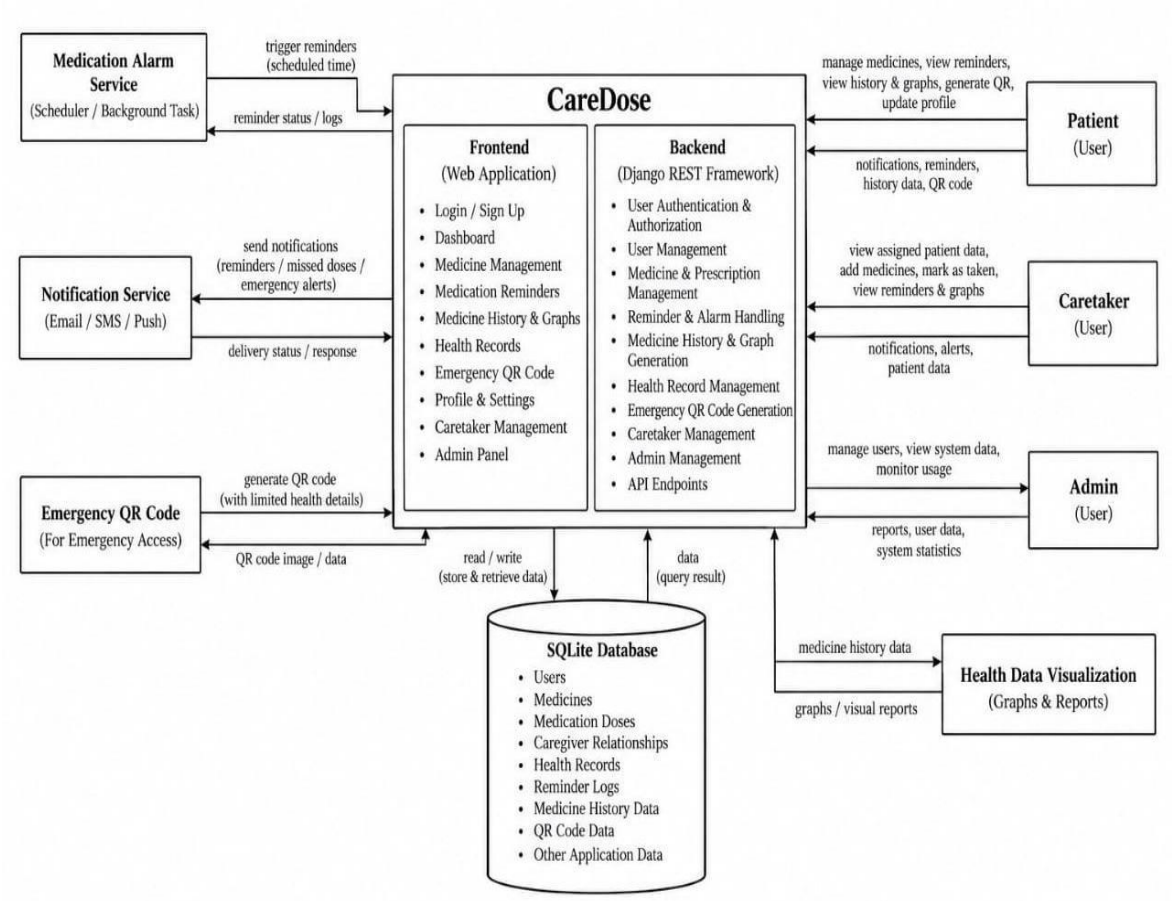


Figure 5.3.1 System Architecture

Sl. No.	Entity	Main Attributes
1	User	userId, name, email, phone, password, role
2	Patient	patientId, dateOfBirth, gender, emergencyContact, weight, pulse
3	Caretaker	caretakerId, name, phone, email, relation
4	Medicine	medicineId, name, dosage, frequency, startDate, endDate, instructions
5	MedicationDose	doseId, date, time, status, takenAt
6	HealthRecord	recordId, condition, allergy, weight, pulse
7	EmergencyQR	qrId, emergencyContact, allergies, currentMedicines, generatedAt
8	Notification	notificationId, type, message, createdAt, read

5.2 Main Database Entities

5.4 Case Study

A representative CareDose case study begins with a patient registering on the platform by providing the required account details and completing their profile with relevant personal and health information. After logging in, the patient can add medicines by entering details such as medicine name, dosage, scheduled timings, frequency, food instructions, treatment duration, and additional notes. The system stores these details and displays the scheduled medicines on the patient dashboard, allowing the user to review upcoming doses and record their medication status as Taken or Pending. The medication history helps the patient review previously recorded dose information and maintain a more organized medication routine.

When additional assistance is required, the patient can establish a connection with a trusted caretaker through the available caretaker functionality. The caretaker can access relevant patient information and medication details according to the permissions provided by the system, supporting family involvement in medication monitoring. CareDose also allows users to maintain health records, including supported measurements such as weight and pulse, and review recorded values through the health-trend dashboard. These graphical representations help users observe changes in their recorded information over time. For emergency situations, the system provides an Emergency Health QR

feature that represents selected emergency information, helping make essential details accessible without exposing the complete health record. In addition, OCR-assisted prescription scanning enables the user to upload a prescription image and extract its text for easier data entry. The extracted information must be checked and corrected by the user before being saved, as OCR may produce inaccurate results. Through these interconnected functions, CareDose demonstrates an integrated approach to medication scheduling, dose tracking, health-record management, caretaker assistance, and controlled emergency information sharing within a single web-based application.

CHAPTER 6

IMPLEMENTATION DETAILS

6.1 Implementation Procedures

The implementation follows a modular procedure in which the user interface, backend API, database, authentication, medication management, caretaker access, health records, OCR, QR generation, dashboard, and administration are developed and integrated incrementally. Each major component is connected to the relevant backend operation and then checked using valid and invalid input cases before integrated testing.

6.1.1 Development Environment

The CareDose project is developed using a standard computer or laptop with a modern web browser and appropriate software development tools. Visual Studio Code is used as the primary code editor for developing and maintaining the application. The frontend is built using HTML, CSS, and JavaScript to provide an interactive and user-friendly interface. The backend is implemented using Python and Django REST Framework (DRF), which provides REST API endpoints for authentication, medicine management, medication tracking, health records, caretaker access, and administrative operations. SQLite is used as the database management system to store and retrieve application data during development and testing. Django's Object-Relational Mapper (ORM) facilitates communication between the backend and the database. Additional supporting tools and services include OCR processing for extracting text from prescription images, QR code generation for emergency health information, chart-rendering functionality for visualizing health records, and the Twilio API for sending SMS medication reminders. These technologies work together to provide the core functionality of the CareDose academic prototype.

Component	Technology Used
Frontend	HTML, CSS, JavaScript
Backend	Python, Django, Django REST Framework
Database	SQLite
Editor	Visual Studio Code
Version Control	Git/GitHub
Browser Testing	Modern Chromium/Firefox-based browser
Charts	JavaScript charting/visualization approach used by the implementation
QR	QR-generation library/component used by the implementation
OCR	OCR component/service integrated into the implementation

6.2 Frontend Implementation

The frontend is organized into separate pages and styles/scripts. The main interfaces include landing/login/signup pages, dashboard, medicine management, history, caretaker access, and administration. CSS is used to provide consistent typography, spacing, colors, cards, forms, and responsive layout. JavaScript handles client-side interaction, validation, dynamic content, and communication with backend services where implemented.

6.3 Backend Implementation

The CareDose backend is implemented using Python and Django REST Framework (DRF), which manages the application's core processing and communication between the frontend and database. REST API endpoints provide structured access to operations related to user authentication, profile management, medicine records, medication tracking, health records, caretaker relationships, and administrative functions. The backend validates incoming requests, verifies user credentials, enforces role-based access permissions, and performs the required database operations using SQLite. It also processes application requests and returns appropriate responses to the frontend in a structured format. The OCR functionality supports the extraction of medicine details from uploaded prescription images, while the Twilio API is integrated to facilitate SMS medication reminders. The backend also supports the application's health-record and emergency information features through the relevant data-handling operations. This separation of responsibilities allows the frontend, developed using HTML, CSS, and JavaScript, to focus on presentation and user interaction, while the backend handles application logic, data processing, and access control.

6.4 Database Implementation

SQLite is used as the database management system for the CareDose academic prototype because it is lightweight, easy to configure, and suitable for development and testing. It provides persistent storage for structured application data, including user accounts, profiles, medicine records, medication schedules, dose statuses, health records, and caretaker relationships. Through Django's Object-Relational Mapper (ORM), the backend can perform database operations using Python models without requiring developers to write SQL queries for every operation. The database helps maintain the relationship between users and their associated records, supporting appropriate account-level data separation. SQLite is suitable for demonstrating the core functionality of CareDose in an academic environment. However, if the application is expanded to support a larger number of users and higher levels of concurrent access, a server-based relational database such as PostgreSQL can be considered to provide better support for scalability and concurrent transactions.

6.5 Authentication and Profile

The authentication system in CareDose manages user registration, login, and access to account-specific information. During registration, the system validates the required fields before creating a new user account. During login, the backend verifies the supplied credentials and grants access only when authentication is successful. Each user's profile information and associated records are linked to the corresponding account to maintain proper data separation. The system also uses role-based access control to direct patients, caretakers, and administrators to the appropriate interfaces and restrict access to unauthorized functions. Passwords are handled through Django's built-in authentication mechanism, which uses secure password hashing rather than storing passwords in plaintext. These measures help protect user credentials, maintain account privacy, and prevent unauthorized access to personal and health-related information.
Add CareDose authentication detailsClarify caretaker and administrator access

6.6 Medicine and Prescription Management

The Medicine Management module enables users to create, view, update, and delete medicine records through a structured interface. Each medicine record can include details such as the medicine name, dosage, scheduled time, frequency, duration, food timing, and additional notes. The system stores these details in the database and associates them with the appropriate user account. Users can review

their saved medicines in an organized list or dashboard, making it easier to access medication information and follow their schedules. The editing functionality allows users to modify existing medicine details when necessary, while the deletion option helps remove records that are no longer required. The stored scheduling information also supports the medication tracking and reminder features of CareDose, including SMS reminders through the Twilio API. Overall, this module provides a centralized way to maintain medicine information and supports organized medication management.

Clarify the medicine scheduling workflowMention user validation and record ownership

6.7 Reminder and Medication Tracking

The Reminder and Medication Tracking module organizes scheduled medicines and helps users monitor their medication routines. Based on the stored medicine details, the system displays relevant medicines and their schedules on the dashboard. Users can update the status of a scheduled dose to Taken, while Pending indicates that the dose has not yet been marked as completed. These status updates are maintained in the medication history, allowing users to review previously recorded medication events. The module also integrates the Twilio API to support sending SMS reminders to the configured phone number at scheduled medication times. The delivery of these reminders depends on the backend scheduling mechanism, Twilio configuration, and network availability. Since CareDose is a web-based academic prototype, continuous background execution and notification delivery may depend on the deployment environment and the scheduling mechanism used. Future enhancements could include browser push notifications or mobile push-notification services such as Firebase Cloud Messaging (FCM) to improve notification support when the application is not actively open.

Clarify how scheduled doses are generatedExplain the reminder scheduling mechanism

6.8 Health Record Management

The Health Record Management module allows users to maintain personal health information separately from their medicine records. It supports recording and retrieving selected health measurements, such as weight and pulse rate, along with their corresponding dates. These records are stored in the database and associated with the appropriate user account, allowing measurements to be maintained and reviewed over time. The dashboard retrieves the available health records and presents the recorded values through graphical representations, making it easier for users to compare previous measurements and observe general trends. Separating health records from medicine records helps

organize different categories of information and simplifies their retrieval for visualization. This module supports personal health monitoring and record keeping rather than medical diagnosis or treatment recommendations.

6.9 Caretaker Module

The Caretaker module is designed to support trusted assistance in managing a patient's medication routine and health information. It allows a patient to establish a relationship with a caretaker, enabling the caretaker to access permitted information associated with the linked patient account. The caretaker interface is intended to support the viewing of medicine schedules and monitoring of medication status. Where the corresponding functionality is implemented, the caretaker can record whether a scheduled dose has been taken or remains pending, helping maintain an updated record of medication activities. Any supported status changes should be reflected in the patient's medication history to ensure consistency between the two interfaces. Access to patient information and available operations is governed by the permissions implemented in the system, preventing unrestricted access to unrelated accounts or administrative functions. This module aims to improve family support and medication monitoring while maintaining appropriate access control and patient data privacy.

6.10 Emergency Health QR

The Emergency Health QR module generates a QR code representing selected health information for convenient access during emergencies. The module is designed to provide essential details without exposing the user's complete health profile by default. Depending on the information configured for emergency access, the QR code may include the user's name, important allergies, major medical conditions, emergency contact number, and other explicitly selected details. When the QR code is scanned using a compatible device, the encoded information can be viewed in a readable format, helping others identify relevant emergency information when the user is unable to communicate it independently. The information included in the QR code should be limited to the fields intended for emergency use, supporting privacy and reducing unnecessary exposure of personal health data. Future improvements may include authenticated access, time-limited QR codes, and additional security measures to prevent unauthorized access. A screenshot of the generated Emergency Health QR screen should be included in this section, along with a screenshot of the scanned output showing the limited emergency information, if available in the final implementation.

6.11 OCR Support

The OCR (Optical Character Recognition) module in CareDose is designed to simplify medicine entry by extracting readable text from uploaded prescription or medicine images. When a user uploads an image, the OCR process identifies available text, which may include medicine names, dosage details, and medication schedules. The extracted information is presented to the user for review and correction before the final medicine record is saved. This process helps reduce manual data entry and improves convenience when adding medicine details to the application. However, the accuracy of text extraction depends on factors such as image quality, text clarity, and handwriting readability. Incorrect or incomplete results may occur, particularly with handwritten prescriptions or unclear images. Therefore, the OCR feature serves only as an assistance tool and does not automatically determine or recommend medication or dosage. Users must verify the extracted details against the original prescription before saving them. This approach helps maintain user control over medicine records while making prescription information entry more convenient.

6.12 Dashboard and Health Trends

The CareDose dashboard serves as the main interface for accessing medication schedules and selected health information. The “Today's Medicines” section displays medicines relevant to the current day based on their configured schedules, helping users identify doses that are due and monitor their medication status. Complete medicine records remain accessible through the medicine management and history pages, allowing users to review saved details and previously recorded medication events. The dashboard also supports the visual presentation of selected health measurements, such as weight and pulse rate, through graphical representations. These charts allow users to compare recorded values across different dates and observe general trends in their health information. By combining medication summaries, dose tracking, and health visualizations in one interface, the dashboard improves access to important information and supports convenient daily medication management and personal health record keeping.

6.13 Admin Module

The Administrator module provides an interface for system-level management and authorized administrative operations within the CareDose application. The administrator is maintained as a separate user role and is not created through the standard patient registration process. Access to the

administrative interface is restricted to authenticated users with the appropriate permissions defined in the backend. Depending on the implemented functionality, the administrator can manage approved system records, user accounts, and relevant application configurations. Administrative privileges are separated from routine patient activities to maintain a clear distinction between system management and personal health record management. The interface is designed to limit unnecessary modification of patient profiles, medicine records, and sensitive health information, helping protect data integrity and privacy. The exact operations available to the administrator depend on the permissions and controls implemented in the backend. This role-based approach supports organized system administration while ensuring that access to user information and administrative functions remains appropriately controlled.

6.14 Testing Methods and Results

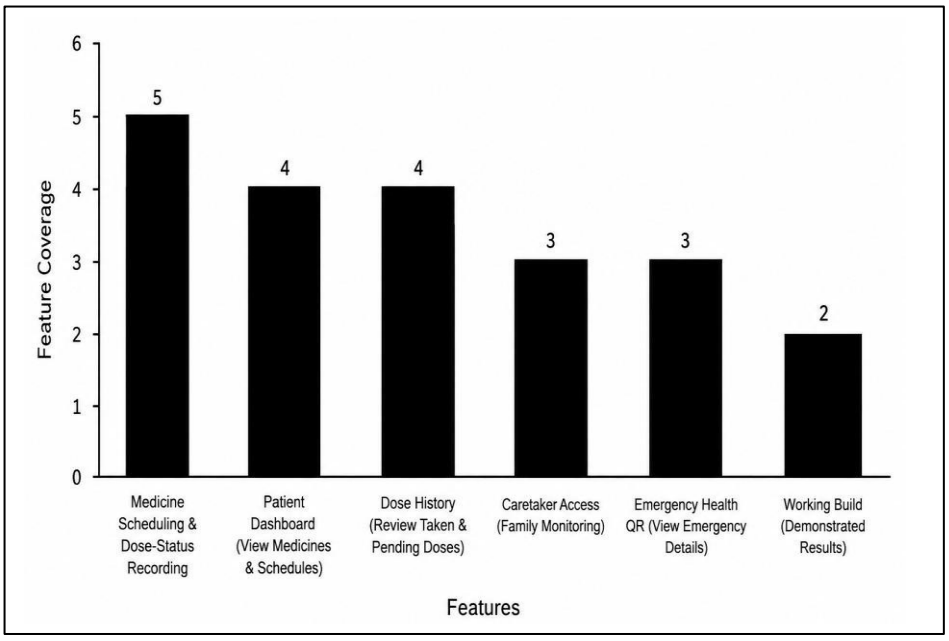
CareDose was tested using different software testing methods such as unit testing, integration testing, functional testing, system testing, and usability testing to ensure that the application works correctly and reliably. Unit testing was performed on individual modules such as user registration, login, profile management, medicine management, medication reminders, and health record management.

Integration testing was conducted to verify the proper interaction between the frontend, Django REST Framework, and SQLite database. Functional testing was used to check important features such as adding and updating medicines, setting medication alarms, marking medicines as taken or missed, viewing medication history, generating the emergency QR code, displaying health graphs, and providing caretaker access.

System testing was performed on the complete CareDose application to ensure that all modules work together without affecting each other. Usability testing was also carried out to check whether the application is simple, clear, and easy to use for patients and caretakers. The testing results showed that the major functionalities of CareDose operated successfully as expected.

User authentication and profile management worked correctly, medicine records were stored and retrieved properly, medication reminders and tracking functions performed as intended, and health information and emergency QR features were displayed correctly. The testing also helped identify and fix errors related to data handling, login, medicine records, and user access. Overall, the results confirmed that CareDose is a functional, reliable, user-friendly, and effective medicine management system.

Feature	**Result**	**Evidence to show**
Medicine schedule	Medicine n timings can be viewed on dashboard	Open patient dashboard
Dose status & History	Taken n pending status can be reviewed	Shows available history
Caretaker Access	Supports patients monitoring	Open the linked patient view
Emergency QR	Displays selected emergency details for quick access	Scan/open QR ,verify details



CHAPTER 7

ADVANTAGES & DISADVANTAGES

7.1 Advantages

- **Centralized Medication Management**

  Medicine details, schedules, status, and history can be managed in one platform.

- **Personalized User Profile**

  The user can maintain selected personal and health information in an account-specific profile.

- **Medication History**

  Taken/pending events can be recorded so that medication activity can be reviewed later.

- **Caretaker Support**

  A trusted person can assist with permitted medication-related activities.

- **Controlled Emergency Information**

  The QR concept provides a compact way to access selected emergency information.

- **OCR-Assisted Entry**

  OCR can reduce repetitive manual entry from prescription or medicine images.

- **Health Trend Visualization**

  Recorded measurements can be represented graphically for easier personal review.

- **Modular Architecture**

  Frontend, backend, database, and individual features can be developed and maintained separately.

- **Account-Level Data Separation**

  Medicine records are associated with the authenticated user, reducing accidental cross-user display.

7.2 Disadvantages

- **Internet/Browser Dependency**
  A web-based system may depend on network connectivity and browser availability for online features.
- **User-Provided Data**
  The quality of medication and health records depends on the accuracy of information entered or confirmed by the user.
- **Privacy and Security Requirements**
  Health information is sensitive and requires strong authentication, authorization, secure storage, and secure communication.
- **OCR Accuracy Limitations**
  OCR can misread unclear images, handwriting, abbreviations, or similar medicine names.
- **Reminder Limitations in a Prototype**
  Reliable background notifications may require platform-specific services or a mobile application.
- **QR Privacy Risk**
  If emergency information is exposed without access control, unauthorized persons could view sensitive details.
- **Not a Medical Diagnostic System**
  The application organizes information and reminders; it does not replace professional diagnosis or treatment.
- **Technical Maintenance**
  The backend, database, authentication, QR, OCR, and notification components require maintenance and security updates.

CHAPTER 8

RESULT & DISCUSSION

The CareDose prototype is evaluated by examining the functionality, usability, and reliability of its implemented modules. The evaluation focuses on the user interfaces, account registration and login, medicine management operations, medication scheduling, dose-status tracking, caretaker access, health-record management, and graphical visualization of recorded health data.

It also examines the OCR functionality for extracting medicine details from uploaded prescription images and the Emergency Health QR feature for displaying selected health information. The Twilio integration is considered as part of the medication reminder functionality, where SMS notifications are used to remind users about scheduled medicines.

Account-level data handling is reviewed to ensure that user information and medicine records are associated with the appropriate accounts. The evaluation is based on the features implemented and their observed behaviour during testing. Screenshots of the final working application, including the major interfaces and key functional outputs, should be included in this chapter to demonstrate the implementation and provide visual evidence of the system's functionality.

8.1 System Output Overview

The CareDose system provides separate interfaces for patients, caretakers, and administrators to support their respective responsibilities.

The patient dashboard serves as the main interface, providing access to medicine schedules, medication reminders, health records, and other health-support features. The medicine management page allows users to add, view, edit, and delete medicine details, including dosage, timing, frequency, and duration. The history page enables users to review recorded medication events and track dose statuses over time

The caretaker interface supports permitted access to linked patient information and helps caretakers monitor medication schedules and provide assistance where required. The system also integrates Twilio for sending SMS medication reminders to support timely medicine intake.

The administrator interface provides system-level controls for managing the platform and overseeing administrative functions. Each interface is designed according to its intended role, helping maintain a clear separation of responsibilities and appropriate access to system features.

8.2 Authentication Results

The authentication workflow is designed to ensure that users can access the CareDose system only through valid registration and login procedures. During registration, the system validates the entered information before creating a new account.

After successful authentication, the user is redirected to the appropriate dashboard based on their assigned role, such as patient, caretaker, or administrator.

If invalid credentials are provided, the system denies access and displays an appropriate error message. Another important aspect of authentication is account-level data separation, which ensures that each user's profile information and medicine records are associated with the correct account.

A newly registered user should see only their own data rather than medicines or profile details belonging to an existing user. These checks help improve access control, protect user information, and ensure that the system maintains appropriate separation between individual accounts.

8.3 Medicine Management Results

The medicine-management feature allows users to create, view, edit, and delete medicine records through a structured interface.

Each record can contain details such as medicine name, dosage, timing, frequency, duration, food instructions, and additional notes. The stored schedule information helps the system organize medicines according to their prescribed timings and identify doses that are due on a particular day.

Users can review their medicine list and update existing records whenever necessary. The medication tracking interface also allows users to record dose statuses, such as Taken or Pending, where supported by the implementation.

These operations help maintain organized medicine records and support regular medication monitoring. The final screenshots included in this section should demonstrate the medicine list, the add and edit medicine forms, and the available dose-status controls to provide visual evidence of the implemented functionality.

8.4 Reminder and Tracking Results

The medication tracking workflow enables users to monitor their scheduled medicine doses by recording their status as Pending or Taken. The dashboard organizes medicines according to their schedules and highlights those relevant to the current day, helping users identify upcoming or due doses.

When a user updates a dose status, the system retains the recorded information in the medication history for future reference. The history page provides a longer-term view of medication events, allowing users to review previous records and monitor their medication-taking patterns over time.

This functionality supports better organization of medicine schedules and helps users maintain a record of their medication intake. The Twilio integration supports the reminder process by sending SMS notifications for scheduled medicines, helping users remember their medication timings. If additional states such as Missed or Snoozed are implemented in the final build, their functionality should be evaluated separately, with corresponding screenshots and test results included in this section. Screenshots of the dashboard, dose-status controls, medication history, and SMS reminders should be provided to demonstrate the implemented tracking and notification workflow.

8.5 Caretaker Interaction Results

The Caretaker module is designed to provide controlled access to selected patient information, enabling trusted individuals to assist patients with medication management. The caretaker can view information made available through the system for the linked patient, while access to unrelated accounts and restricted administrative functions remains limited. This approach helps maintain appropriate access control and protects the privacy of patient records.

Where medication-status controls are supported, the caretaker can update the relevant dose status, such as Taken or Pending, to assist in monitoring the patient's medication routine. The recorded changes should be reflected in the patient's medication history to maintain consistency between the patient and caretaker interfaces. This functionality can be particularly useful for elderly patients or individuals who require assistance in following their medication schedules.

The evaluation should verify that the caretaker can access only permitted information and that any supported medication-status updates are correctly recorded. Screenshots of the caretaker interface, patient information view, and medication-status updates should be included to demonstrate the implemented functionality.

8.6 Health Record and Trend Results

The Health Record Management and Visualization module is evaluated by checking whether users can enter, save, retrieve, and review their health information through the profile or health section of the CareDose application. The module maintains selected health records, such as weight and pulse rate, allowing users to review previously recorded measurements.

The visualization component presents available health data through graphical representations, plotting recorded values against their corresponding dates to make changes over time easier to observe.

The evaluation focuses on the correctness of data storage and retrieval, the accuracy of the displayed values, and the proper representation of records in the charts.

These visualizations help users organize their health information, compare previous measurements, and observe general trends.

The module is intended for personal health monitoring and record keeping and does not provide medical diagnosis or replace professional medical advice.

Screenshots of the health-record entry interface, saved records, and corresponding graphs should be included to demonstrate the implemented functionality.

8.7 OCR and QR Results

The OCR and Emergency Health QR features are evaluated to assess their functionality, accuracy, and usefulness within the CareDose system. The OCR feature is tested using clear and unclear sample images of medicine labels or prescriptions to observe how effectively the system extracts readable information, such as medicine names, dosage details, and medication schedules.

The extracted text is reviewed to ensure that users can correct any inaccurate or incomplete information before saving the final medicine record. This evaluation helps determine whether the OCR functionality reduces manual data entry while allowing users to verify the correctness of the extracted details.

The Emergency Health QR feature is evaluated by generating a QR code containing selected patient health information and scanning it using a compatible device. The displayed information is checked to ensure that it includes only the intended emergency details, such as the patient's name, known allergies, major medical conditions, and emergency contact information, according to the configured implementation.

The evaluation also considers whether the QR code can be scanned and whether the information is presented in a clear and readable format.

These tests help assess the practical usefulness of both features in simplifying medicine entry and providing convenient access to essential emergency information.

Screenshots of the OCR extraction and correction process, along with the generated QR code and its scanned output, should be included as evidence of the implemented functionality.

8.8 Notification and SMS

The Notification and SMS module is evaluated to verify whether medication reminders are triggered according to the schedules configured by the user and whether SMS messages are successfully sent through the Twilio API. The evaluation focuses on the accuracy of reminder timing, the correctness of the message content, and the successful delivery of notifications to the configured phone number. The SMS notification is intended to remind users when it is time to take their scheduled medicines, helping them follow their medication routines more consistently. The system's behaviour is also examined when notification delivery fails due to network issues, invalid phone numbers, or Twilio configuration errors.

Since SMS delivery may depend on network availability and account restrictions, successful message submission does not always guarantee immediate delivery. Screenshots of the reminder configuration, Twilio console, and received SMS messages should be included as evidence of the implemented functionality.

Notification response times and delivery success rates should be reported only after actual testin

8.9 System Performance and Discussion

For The CareDose prototype uses SQLite as its database and Django REST Framework (DRF) to support backend operations and communication between the frontend and backend.

For the current academic prototype, the main Create, Read, Update, and Delete (CRUD) operations are designed to handle medicine records, user profiles, medication tracking, and health information. The lightweight database setup is suitable for developing and testing the core functionality of the application. However, several performance considerations must be addressed before deploying the system on a larger scale.

These include database scalability, concurrent user access, reliable delivery of SMS notifications through Twilio, processing time for uploaded prescription images during OCR operations, secure access to emergency information through QR codes, and protection of health data during transmission.

Future improvements may include database optimization, stronger access controls, secure communication using HTTPS, and more efficient handling of image-processing tasks. Formal performance testing should be conducted to measure response times, resource usage, and system behaviour under different workloads.

Quantitative performance results should be included only after actual measurements have been obtained, rather than using estimated values without supporting test data.

CHAPTER 9

CONCLUSION

9.1 Summary

CareDose: Smart Medicine Reminder & Health Companion is a web-based application designed to simplify medication management by bringing medicine schedules, dose tracking, personal health records, caretaker support, and emergency health information together in a single platform. The system aims to reduce the difficulties associated with managing medication details through scattered prescriptions, handwritten notes, memory, phone alarms, and separate applications. It provides a centralized platform where patients can manage their medicines, view scheduled reminders, record taken and pending doses, and review their medication history. The application also incorporates OCR-assisted prescription scanning to extract text from prescription images, helping reduce manual data entry, while health-trend visualization enables users to review recorded information through graphical representations. In addition, caretaker support facilitates family involvement in medication monitoring, and the Emergency Health QR feature provides controlled access to selected health information when required. CareDose is developed using HTML, CSS, and JavaScript for the frontend, with Python, Django REST Framework, and SQLite supporting backend processing and data storage. Twilio is integrated to support SMS reminders, subject to account restrictions, while an automated scheduler checks medicine timings and triggers reminder messages. The system follows a modular design to organize its different functionalities and incorporates authentication and role-based access to help protect user information. Overall, CareDose demonstrates how commonly available web technologies can be combined to develop an accessible and organized medication-support application. It also emphasizes the importance of privacy, responsible health-information sharing, and user verification, while serving as a supportive tool rather than a replacement for professional medical advice or treatment.

9.2 Limitations

- The current prototype may depend on browser and network availability for online functions.
- Reliable background notifications may require a dedicated notification service or mobile application.
- OCR results can be inaccurate and require user verification.
- Health data is dependent on correct user input.
- Emergency QR access must be carefully secured to prevent unauthorized exposure.

- The current academic database is SQLite and may not be suitable for large-scale production deployment without architectural changes.
- The system does not replace professional medical advice, diagnosis, prescription, or emergency care.
- Advanced drug-interaction checking, if not included in the final implementation, remains a future enhancement rather than a current feature.
- Third-party health-platform integration is not assumed unless it is explicitly implemented and tested.

9.3 Future Scope

- Mobile application development for Android and iOS.
- Reliable push notifications and recurring background reminders using an appropriate notification service.
- Integration with Samsung Health, Google Health Connect, or other approved health platforms where APIs and permissions permit.
- Time-limited or permission-based emergency QR access.
- Secure sharing of selected health records with doctors or hospitals.
- Advanced drug-interaction and medication-conflict checking using validated medical data sources.
- Improved OCR with medicine-name verification and structured prescription parsing.
- Multilingual support for wider accessibility.
- Voice-based reminders and accessibility features.
- Cloud deployment with a production-grade relational database.
- Audit logs for caretaker and administrative actions.
- Analytics for medication adherence patterns based on user-recorded data.
- Optional pharmacy/refill reminders and prescription renewal tracking.

9.4 Applications

- Personal medication management: Helps users organize medicine names, dosages, schedules, food instructions, and treatment durations in one place for convenient access and tracking.

- Medication support for elderly users: Supports elderly patients who may need assistance remembering medicine timings and maintaining a regular medication routine through scheduled reminders and dose-status tracking.
- Family and caretaker-assisted medication tracking: Enables caretakers to support linked patients by accessing relevant medication information and monitoring their medication routines, subject to the system's access permissions.
- Health-record organization: Provides a centralized place to maintain personal health details, relevant medical information, and medication history, making it easier for users to review their recorded information.
- Emergency health-information sharing: Provides an Emergency Health QR feature that allows selected emergency details to be accessed quickly when required, while limiting the information exposed.
- OCR-assisted prescription management: Helps extract text from uploaded prescription images using Optical Character Recognition (OCR), reducing manual entry effort while allowing users to verify the extracted details.
- Health-trend visualization: Presents recorded medication and health-related information through graphical representations, helping users review historical records and observe patterns over time.
- SMS-based medication reminders: Uses Twilio and an automated scheduler to trigger SMS reminders according to configured medicine schedules when the scheduler is running and the messaging service is available.
- Academic demonstration of healthcare web applications: Demonstrates the practical use of HTML, CSS, JavaScript, Python, Django REST Framework, and SQLite to develop a modular healthcare-support application.

Future integration with digital health ecosystems: Provides a foundation for potential future enhancements, such as integration with wearable devices, digital health platforms, advanced medication analytics, and additional notification services.

CHAPTER 10

APPENDICES

Login page

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>CareDose - Login</title>
<link rel="stylesheet" href="css/style.css">
</head>
<body>
<div class="login-page">
<div class="login-box">
<h1>CareDose</h1>
<p>Your Smart Medicine Reminder</p>
<form id="loginForm" onsubmit="handleLoginSubmit(event)">
<label for="loginEmail">Email Address</label>
<input type="email" id="loginEmail" placeholder="Enter your email" required>
<label for="loginPassword">Password</label>
<input type="password" id="loginPassword" placeholder="Enter your password" required>
<button type="submit">Login</button>
<p>Don't have an account? <a href="signup.html">Sign Up</a></p>
</form>
</div>
</div>
<script src="js/script.js"></script>
</body>
</html>
```

Signup Page

```
<!DOCTYPE html>
```

```
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>CareDose - Sign Up</title>
<link rel="stylesheet" href="css/style.css">
</head>
<body>
<div class="signup-container">
<h1>CareDose</h1>
<p>Create your health management account</p>
<form id="signupForm" onsubmit="handleSignupSubmit(event)">
<label for="signupName">Full Name</label>
<input type="text" id="signupName" placeholder="Full Name" required>
<label for="signupEmail">Email Address</label>
<input type="email" id="signupEmail" placeholder="Email Address" required>
<label for="signupPhone">Phone Number</label>
<input type="tel" id="signupPhone" placeholder="Phone Number" maxlength="10" required>
<label for="signupPassword">Password</label>
<input type="password" id="signupPassword" placeholder="Password" minlength="8" required>
<label for="confirmPassword">Confirm Password</label>
<input type="password" id="confirmPassword" placeholder="Confirm Password" required>
<button type="submit">Create Account</button>
<p>Already have an account? <a href="login.html">Sign In</a></p>
</form>
</div>
<script src="js/script.js"></script>
</body>
</html>
```

Profile page

```
<html>
<div class="profile-page">
<h2>My Profile</h2>
```

```
<form id="profileForm">
<input type="text" id="name" placeholder="Full Name" required>
<input type="number" id="age" placeholder="Age">
<input type="text" id="bloodGroup" placeholder="Blood Group">
<input type="tel" id="emergencyContact" placeholder="Emergency Contact" maxlength="10" required>
<input type="text" id="allergies" placeholder="Allergies">
<button type="submit">Save Profile</button>
</form>
</div>
<script>
document.getElementById('profileForm').addEventListener('submit', function(e) {
e.preventDefault();
const profile = {
name: document.getElementById('name').value,
age: document.getElementById('age').value,
bloodGroup: document.getElementById('bloodGroup').value,
emergencyContact: document.getElementById('emergencyContact').value,
allergies: document.getElementById('allergies').value
};
localStorage.setItem('profile', JSON.stringify(profile));
alert('Profile saved successfully!');
});
</script>
</html>
```

Medicine page

```
def add_medicine(request):
if request.method == "POST":
name = request.POST.get("name")
dosage = request.POST.get("dosage")
frequency = request.POST.get("frequency")
start_date = request.POST.get("start_date")
```

```
times = request.POST.getlist("times")

if name and dosage and start_date:
Medicine.objects.create(
patient=request.user,
name=name,
dosage=dosage,
frequency=frequency,
start_date=start_date,
times=times
)
return redirect("medicine_list")

return render(request, "add_medicine.html")
```

OCR

```
import pytesseract
from PIL import Image
from django.shortcuts import render

def scan_prescription(request):
extracted_text = ""
if request.method == "POST" and request.FILES.get("prescription"):
image_file = request.FILES["prescription"]
image = Image.open(image_file).convert("RGB")

extracted_text = pytesseract.image_to_string(image)

return render(request, "prescription.html", {
"extracted_text": extracted_text,
})
return render(request, "scan.html", {
"extracted_text": extracted_text,
})
```

Notification

```
from plyer import notification
from datetime import datetime
import time

def medicine_notification(medicine, dosage):
notification.notify(
title="CareDose - Medicine Reminder",
message=f"It's time to take {medicine} ({dosage}).",
app_name="CareDose",
timeout=10
)

while True:
current_time = datetime.now().strftime("%H:%M")
if current_time == "20:00":
medicine_notification("Aspirin", "1 tablet")
time.sleep(60)
time.sleep(1)
```

Appendix C – Important Implementation Notes

- User data isolation: Ensure that each authenticated user can access and manage only their own data, using the authenticated user ID to prevent unauthorized access to another patient's information.
- Emergency QR privacy: Avoid exposing complete health records through an unrestricted QR code. Include only selected emergency information and apply appropriate access controls wherever possible.

- Password security: Never store passwords in plaintext. Use Django's password-hashing mechanisms to protect user credentials.
- Input validation: Validate medicine details, profile information, and other user inputs at both the frontend and backend wherever applicable to reduce invalid data and improve security.
- OCR verification: Treat text extracted from prescription images as unverified information. Allow users to review and correct medicine names, dosages, and other details before saving them.
- Role-based access control: Keep caretaker permissions limited to the information and actions required for patient assistance, while administrative permissions are reserved for authorized system-management activities.
- Medical responsibility: Clearly present CareDose as a medication-management and informational support tool, not as a substitute for professional medical diagnosis, prescriptions, or treatment.
- Accurate testing records: Record actual test results only after executing the corresponding test cases. Clearly distinguish successfully tested features from features that are planned or not yet verified.

CHAPTER 11

REFERENCES

1. International Journal of Engineering Development and Research (IJEDR), "Smart Medicine Reminder System Using OCR with Family Monitoring," Vol. 14, Issue 2, April 2026, ISSN 2321-9939.

2. L. C. Armitage, A. Kassavou, and S. Sutton, "Do mobile device apps designed to support medication adherence demonstrate efficacy? A systematic review of randomised controlled trials, with meta-analysis," BMJ Open, vol. 10, no. 1, e032045, 2020, doi:10.1136/bmjopen-2019-032045.

3. V. Pérez-Jover, M. Sala-González, M. Guilabert, and J. J. Mira, "Mobile Apps for Increasing Treatment Adherence: Systematic Review," Journal of Medical Internet Research, vol. 21, no. 6, e12505, 2019, doi:10.2196/12505.

4. K. Santo, S. S. Richtering, J. Chalmers, A. Thiagalingam, C. K. Chow, and J. Redfern, "Mobile Phone Apps to Improve Medication Adherence: A Systematic Stepwise Process to Identify High-Quality Apps," JMIR mHealth and uHealth, vol. 4, no. 4, e132, 2016, doi:10.2196/mhealth.6742.

5. M. Vervloet, A. J. Linn, J. C. M. van Weert, D. H. de Bakker, M. L. Bouvy, and L. van Dijk, "The effectiveness of interventions using electronic reminders to improve adherence to chronic medication: a systematic review of the literature," Journal of the American Medical Informatics Association, vol. 19, no. 5, pp. 696–704, 2012, doi:10.1136/amiajnl-2011-000748.

6. R. E. Sarabi, F. Sadoughi, R. J. Orak, and K. Bahaadinbeigy, "The Effectiveness of Mobile Phone Text Messaging in Improving Medication Adherence for Patients with Chronic Diseases: A Systematic Review," Iranian Red Crescent Medical Journal, vol. 18, no. 5, e25183, 2016, doi:10.5812/ircmj.25183.

7. B. P. H. Pouls, J. E. Vriezekolk, C. L. Bekker, A. J. Linn, H. A. W. van Onzenoort, M. Vervloet, S. van Dulmen, and B. J. F. van den Bemt, "Effect of Interactive eHealth Interventions on Improving Medication Adherence in Adults With Long-Term Medication: Systematic Review," Journal of Medical Internet Research, vol. 23, no. 1, e18901, 2021, doi:10.2196/18901.

8. M. Paterson, M. Kinnear, C. Bond, and B. McKinstry, "A systematic review of electronic multi-compartment medication devices with reminder systems for improving adherence to self-administered medications," International Journal of Pharmacy Practice, vol. 25, no. 3, pp. 185–194, 2017, doi:10.1111/ijpp.12242.

9. M. Espírito-Santo, S. Santos, and M. D. Estêvão, "Digital Solutions Available to Be Used by Informal Caregivers, Contributing to Medication Adherence: A Scoping Review," Pharmacy, vol. 12, no. 1, p. 20, 2024, doi:10.3390/pharmacy12010020.

10. “Image-Based Extraction of Prescription Information using OCR-Tesseract,” Procedia Computer Science, vol. 235, pp. 1077–1086, 2024, doi:10.1016/j.procs.2024.04.102.

11. World Health Organization, Adherence to Long-Term Therapies: Evidence for Action. Geneva: World Health Organization, 2003.

12. L. Osterberg and T. Blaschke, “Adherence to Medication,” New England Journal of Medicine, vol. 353, no. 5, pp. 487–497, 2005, doi:10.1056/NEJMra050100.

13. B. Jimmy and J. Jose, “Patient Medication Adherence: Measures in Daily Practice,” Oman Medical Journal, vol. 26, no. 3, pp. 155–159, 2011, doi:10.5001/omj.2011.38.

14. M. T. Brown and J. K. Bussell, “Medication adherence: WHO cares?” Mayo Clinic Proceedings, vol. 86, no. 4, pp. 304–314, 2011, doi:10.4065/mcp.2010.0575.

15. D. S. Wald, S. Butt, and J. P. Bestwick, “One-way versus two-way text messaging on improving medication adherence: meta-analysis of randomized trials,” American Journal of Medicine, vol. 128, no. 10, pp. 1139.e1–1139.e5, 2015, doi:10.1016/j.amjmed.2015.05.035.

16. P. Kardas, P. Lewek, and M. Matyjaszczyk, “Determinants of patient adherence: a review of systematic reviews,” Frontiers in Pharmacology, vol. 4, article 91, 2013, doi:10.3389/fphar.2013.00091.

17. World Health Organization, “Medication Safety” and digital health resources, official guidance and technical materials.

18. Django Software Foundation, Django Documentation, official framework documentation.

19. Django REST Framework, official documentation for REST API development with Django.

20. Python Software Foundation, Python Documentation, official language and standard-library documentation.

21. SQLite Consortium, SQLite Documentation, official database and SQL documentation.

22. Mozilla Developer Network (MDN), HTML, CSS, JavaScript and Web APIs documentation.

23. Google, Firebase Cloud Messaging Documentation, notification and messaging reference.

24. Android Developers, Health Connect Documentation, official Android health-data integration reference.

25. Samsung, Samsung Health / Samsung Health SDK documentation, subject to platform availability and permissions.

Gayathri Krishnan

mca_final(final)

mini project report

Document Details

Submission ID

trn:oid:::10159:154548995

Submission Date

Oct 13, 2026, 5:03 PM GMT+5:30

Download Date

Oct 13, 2026, 5:08 PM GMT+5:30

File Name

mca_final(final).docx

File Size

4.6 MB

76 Pages

14,424 Words

92,989 Characters

15% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text
- Small Matches (less than 8 words)

Exclusions

- 8 Excluded Matches

Match Groups

176 Not Cited or Quoted 15%
Matches with neither in-text citation nor quotation marks

0 Missing Quotations 0%
Matches that are still very similar to source material

0 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

9% Internet sources
4% Publications
13% Submitted works (Student Papers)